

Elmo Gold Drive Firmware Release Notes

Version: 01.02.18.00 B00
For Elmo Gold Drives

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Chapter 1. General

The following Gold Level release notes include all modifications from version 01.01.07.15 released in January 2014 until version 01.02.18.00 B00 released in February 2024.

Note

Modifications between firmware versions may result in parameter database going back to default values. It is advisable to upload the drive's parameters via the Elmo Application Studio (EAS) prior of updating any firmware version. The parameters can be downloaded to the drive after the firmware was downloaded.



Chapter 2. 01.02.18.00 B00

Component	Version
Firmware Version	01.02.18.00 13Feb2024B00
Communication Server	1.1.9.07
PAL	90
ESI	0x00010420 V14

2.1 Details

Subject	Details
New Profile Position target processed in Quick stop	From this version, new Set Points that the drive received via Object 0x607A are processed in the <i>Quick Stop</i> state as well. The outcome of the modification is that the motor will move to the new Set Point as soon as the state machine will be switched back to <i>Operational Enable</i> state. Previously, Set Points that were received during the <i>Quick Stop</i> state were ignored.
New homing mode under Halt	From this version, new DS402 Homing method that are requested when the drive is in Halt are processed and executed as soon as the “resume to motion” is permitted. Previously new Home methods were ignored.
Tuba 30/480 High PWM switching	From this version Tuba 30/480 can be activated in PWM frequency of 20KHz via the XP[2] command. Note that with this frequency the max current is limited to 20 amps.

2.2 Bug fixes 1.2.18.00 B00

Bug	Details
IAI sensor Zero Cross detection when position is inverted	Zero Cross detection emulates Index allowing the application to e.g. perform Homing on Index or count a single turn event. With the IAI sensor, when the Zero Cross was used for Homing on Index, the capture position (Home position) was wrongly calculated when the sensor readout was defined as “Invert” via CA[28] . This was fixed. A correct capture is reported regardless of the “Invert bit” state.



Chapter 3. 01.02.17.05 B00

Component	Version
Firmware Version	01.02.17.05 12Dec2023B00
Communication Server	1.1.9.07
PAL	90
ESI	0x00010420 V14

3.1 Details

Subject	Details
Present Current Command before cogging	The Gold drive supports method to overcome a non-linear cogging disturbances. The current command before the cogging compensation is now reported via WI[49] . The value of the current command can also be mapped to a PDO via object 0x2203 (Application Object) by setting 0x2F41 to 10.
Stepper commutation improvement	The stepper commutation procedure can result in an undesired fast current modification. From this version the current can be gradually decreased per SC[16] . The parameters is in seconds and default is 0.01.

3.2 Bug fixes 1.2.17.05 B00

Bug	Details
Non-Dual-Use speed limit	In some situations the speed for a Q drives was wrongly calculated resulted in lower speed than required. Typically happened when using a high resolution sensor for the commutation. This was fixed. In any case the electrical speed is limited to 599Hz.
CST mode speed calculation	In Cyclic Synchronous Torque mode under Halls Only the speed calculation was wrong resulting in a wrong speed limit protection. This was fixed.
CANOpen NMT reset after motor fault	In some rare situations setting NMT mode 0x81 (reset Node) following a motor faults that occurred during the servo enable procedure (via MO=1 or via DS402 Operation Enable state), resulted in damage to the power bridge. The relevant faults are "Speed too High" and "Halls mismatch". This was fixed.



Bug	Details
Position tracking error under CANopen IP mode	A steady state position error was detected during constant speed when the drive is under CANopen Interpolated Position mode and a high resolution is used. This was fixed.



Chapter 4. 01.02.17.04 B08

Component	Version
Firmware Version	01.02.17.04 22May2023B08
Communication Server	1.1.9.07
PAL	90
ESI	0x00010420 V14

4.1 Details

Subject	Details
IAI sensor	The IAI sensor is supported including the IS & RCS types for 17 & 15 bits of single turn with option for a multi rotation of 552 (-276 ~ +275). Sensor is supported by the Elmo Application Studio as well.

4.2 Bug fixes 1.2.17.04 B08

Bug	Details
Handling SDO size under EtherCAT mailbox read option	If the EtherCAT host request expedite SDO that is longer than the permitted size, the drive copies the maximum permitted size. Previously in such situations the drive fetched a wrong memory allocation and failed. In some situations the failure resulted in a request for another firmware download.
Non-Dual-Use drives – lower speed when different sensor resolution is used.	If different sensors were used for a Non-Dual-Use drives (Q Drives) and the resolution of the commutation sensor differ from the resolution of the velocity sensor, the speed limit preventing the dual use was wrongly calculated. The new CA[124] parameter denote the factor between the velocity and the commutation sensors in order to allow the proper speed. The units of the new parameter are 1/10000 which means that 10000 (default) is '1' which means that the commutation sensor and the velocity sensor have the same resolution.



Chapter 5. 01.02.17.04 B00

Component	Version
Firmware Version	01.02.17.04 13Mar2023B00
Communication Server	1.1.9.05
PAL	90
ESI	0x00010420 V14

5.1 Details

Subject	Details
Jori \ Leopard new power rate	Jori with max current of 70 amps was added. The bus voltage is 800V DC. Drive availability are per marketing announcements only.

5.2 Bug fixes 1.2.17.04 B00

Bug	Details
Wrong Demand value with modulo	Object 0x6062 is the demand value for the Position Control loop. The values may have overshoot when reaching the high limit of the position modulo as denoted via XM[2] . This was fixed.
XP[11] current derating voltage threshold	The XP[11] threshold, if was set by a user, was calculated wrongly and may have started the derating of the current on lower voltage than intended. This was fixed.
FoE download last message	In situation where the file downloaded via FoE have the same size as the FoE mailbox, a last empty message is required to conclude the transfer. Before the fix drive concluded the transfer previously. This was fixed. Drive is waiting for an empty message in such rare situations.
ESI file 10420 V14	Some objects properties were fixed from UINT to INT. Objects 0x6070 & 0x606F mapping attribute was removed Object 0x2E00 map-able attribute was added These modifications effected only the offline dictionary from the ESI. The on-line dictionary from the drive was not modified.



Chapter 6. 01.02.17.02 B00

Component	Version
Firmware Version	01.02.17.02 24Jul2022B00
Communication Server	1.1.9.04
PAL	90
ESI	0x00010420 V12

6.1 Details

Subject	Details
QV[] & QT[] commands	These commands were removed from old versions and now are restored for the sake of general usage. Each entry includes 64 entries, QT is sign 16 bits and QV is signed 32 bits.

6.2 Bug fixes 1.2.17.02 B00

Bug	Details
Current limits initiation	In CST mode the current limits were not initiated which prevent the stepper commutation to be properly activated. This was fixed.



Chapter 7. 01.02.17.01 B00

Component	Version
Firmware Version	01.02.17.01 13Apr2022B00
Communication Server	1.1.9.04
PAL	90
ESI	0x00010420 V12

7.1 Details

Subject	Details
Adding Jori \ Leopard to Gold Family	Jori and Leopard are high power drives based on TWI\BEE. These are “R” type drives with maximum current of 30 & 60 and 800V DC. Note that the drives are available for orders only when announced by marketing and not by this document.

7.2 Bug fixes 1.2.17.01 B00

Bug	Details



Chapter 8. 01.02.17.00 B10

Component	Version
Firmware Version	01.02.17.00 22Mar2022B10
Communication Server	1.1.9.03
PAL	90
ESI	0x00010420 V12

8.1 Details

Subject	Details

8.2 Bug fixes 1.2.17.00 B10

Bug	Details
Downloading between mixed version	An attempt to download version from 1.1.xx.xx to a drive with new communication chip will now results in a failure. Previously the drive could be locked and lose communication for good.
Mitsubishi set up	In some situation the Mitsubishi sensor setup failed and required a reset. This was fixed.
Sine signal with encoder sensor	The sine signal could not be activated with an encoder sensor. This was fixed.



Chapter 9. 01.02.17.00 B01

Component	Version
Firmware Version	01.02.17.00 16Jan2022B01
Communication Server	1.1.9.00
PAL	90
ESI	0x00010420 V12

9.1 Communication Processor Adjustment

Due to a global component shortage, the firmware version of the drive was adjusted to fit a reduced memory chip-set.

The modifications are directly related to the communication processor only and have no effect on the motion, servo, or other operational features of the drive.

The new communication processor revision supports memory of 256K while the previous supported 512K.

The modification includes both boot and firmware versions.

The modified version is relevant to all firmware files allowing the firmware downloading via all communication channels - FoE & USB\RS223.

The modified version can be downloaded seamlessly to both types of drives - 256K or 512K.

All **present firmware versions** support the **512K only**. These versions cannot be downloaded to the 256K drives. An attempt of doing so will result in an immediate error and the drive remains in boot state.

In order to maintain compatibility with existing EtherCAT configuration files and minimize integration time, the product code (**0x1018.2**) and the revision number (**0x1018.3**) were not modified.

These are the modifications:

Subject	Details
Firmware version	The drive firmware versions via VR , AA[5] and object 0x100A are modified to reflect the new change. See the release note.
Telnet Removed	Telnet is no longer available in this version.
DHCP Removed	DHCP is no longer available, static addressing is required.
Unable to update boot loader version via FoE	The ability to update the boot loaded via FoE is no longer supported.
UDP buffers reduced	Under Ethernet mode the UDP buffers are reduced, this may cause some limitations in the situation of broadcast messages if



Subject	Details
	for example, the drives are connected to a factory network while communicating via UDP.

9.2 Details

Subject	Details
Communication processor 1 st version	The version includes the modifications reported on Chapter 2 In this doc
Mitsubishi sensor supported	The Mitsubishi serial sensors with single turn of 14-26 and multi turn of 16. The sensor is supported from EAS version 2.9.

9.3 Bug fixes 1.2.17.00 B01

Bug	Details



Chapter 10. 01.01.16.10 B04

Component	Version
Firmware Version	01.01.16.10 16Dec2021B04
Communication Server	1.1.8.6
PAL	90
ESI	0x00010420 V12

10.1 Details

Subject	Details
NRZ sensors roll over	Indication for absolute sensor roll over is not available for NRZ sensors. The roll over can be used as Index
Emulation position indication	From this version WI[59] indicates the emulation position as calculated by the drive.
Dual Use with DC brush motors	Until this version the Dual use (Q drives) could not be operated with DC brush motors. This was fixed.

10.2 Bug fixes 1.1.16.10 B04

Bug	Details
Smooth Factor - SF not active	The SF command was not effective after power up and required a new setting of the SF[1] each time. This was fixed. The issues started when the SF[2] was added (version 1.1.16.0 B01). This was fixed.
Home attain bit in DS402 Status word	In DS402 Status word under homing mode, bit 12 is set when the home is attained. This bit is set to '1' when the drive is in Homing Mode (0x60601 = 6) and after a successful homing procedure in all homing methods. When the homing method was "Immediate Homing" bit 12 was not reset to '1' as expected when Homing mode was requested again. This was fixed. The bit behaves in "Immediate homing mode" like in all modes.
CSP mode when FLS\RLS limit switches are encountered	When drive under cyclic synchronous position mode and a limit switch was hit, an attempt to disable the drive via control word 6 (shutdown command) failed and the drive remained in servo enable. This situation is relevant when the shutdown option code requires deceleration to 0 speed.



Bug	Details
	The bug was fixed. The drive, in this situation, will be disabled after the velocity command will decelerate to 0.



Chapter 11. 01.01.16.05 B00

Component	Version
Firmware Version	01.01.16.05 04Mar2021B00
Communication Server	1.1.8.6
PAL	90
ESI	0x00010420 V12

11.1 Details

Subject	Details
Twitter and Bee PWM multiplication enhanced	PWM multiplication is done via XP[2] command. From this version the TWI & BEE with BV=100 & MC=20 can have multiplication of x3 (XP[2]=6). Note that the permitted current limits will be reduced to 50% from the nominal one (when the multiplication is 1)
Current phases sum tracking error	The total current derived from the motor phases should be 0. From this version, ER[10] & ER[11] commands can be used to set threshold for differences of the current phases (ER[10]) with time window (ER[11]). The current sum tracking error assures that in situations that the sum of the current exceeds the threshold of ER[10] (in milliamps) for ER[11] (in milliseconds), than the motor is disabled with MF=0x4000000 and EMCY message 0x8382. Note: the parameter database will be reset to default.
Velocity limit in Cyclic Synchronous Torque mode	In CST mode the velocity can be limited to a desired value. This is done by reducing the current limit as follows: The velocity limit is set via object 0x6080 (in RPM). Via XA[10] the user sets the desired limits in 0.1% from 0x6080. For example: if the 0x6080 is 2000 [RPM] and the XA[10] is 800 (i.e. 80%) then the torque will decrease from 1600 [RPM] which are the 80% from 0x6080 to 2000 [RPM] in such a way that in 2000 RPM the torque will be 0.
NF[9]: input shaping frequency limit improved	The frequency of the input shaping are now improved for: NF[8]=1 (ZV mode) limit to 2.02Hz. NF[8]=2 (ZVD mode) limit to 4.05Hz.
Twitter \ Bee new power rates	The TWI and BEE MC=100 BV=100 were added to the Gold family and can be used from this version on.



11.2 Bug fixes 1.1.16.05 B00

Bug	Details
Unplanned final speed motion	Sequence of commands that included JP->SP->PA & BG, resulted in a continuation of the motion with Final Speed (FS) of SP. This was fixed.
Field angle is not reset in UM=3	When motor is enabled in UM=3 mode the field angle (stepper angle) was not reset to 0 resulting in different start of the motor every motor enable. This was fixed. Field angle is 0 in this mode.
Hall mismatch adjusted	In some situation the Halls width were not used properly (CA [115 – 119]) this was fixed.
Motor enable fails after fault	In some rare situations the motor enabled procedure (MO =1) could be completed successfully only after 2 nd attempt. This happened only after an attempt to enable after a motor fault. This was fixed.
NMT buffer overflow EMCY message	The CANopen NMT allows the host to control over the state machine of the CANopen communication. In situations that the buffer was overflowed we got undesired EMCY messages. This was fixed. Only relevant NMT messages are processed by the drive.
CST mode: bit 12 statuword	In CST mode bit 12 of object 0x6041 indicates if the set point is processed ('1') or not ('0'). This indication was not handled properly in Cyclic Synchronous Torque mode. This was fixed.
TCP/IP buffer stacked	Bursts of ARP messages to the drive resulted in situations that messages process were stacked and not processed anymore even after the bursts were stopped. This was fixed.
Resolver capture modes	Using the resolver feedback, capture modes were not available preventing all sorts of homing and position capture modes (e.g. touch probe). This was fixed. All touch modes are now available with resolver as well.



Chapter 12. 01.01.16.00 B01

Component	Version
Firmware Version	01.01.16.00 05Mar2020B01
Communication Server	1.1.8.1
PAL	90
ESI	0x00010420 V11

12.1 Details

Subject	Details
Drum/Eagle merged to official version	DRU\EAG with the following rate are added to the official version: 430/80, 650/80, 360/100, 550/100. The following details are relevant for these drives only.
Current derating in Drum/Eagle due to voltage limits	For the above mentioned drives the maximum current of the drive is now de-rated as a function of the input bus voltage. The De-rating starts at the threshold limit defined by XP[11], and decrease from its max value, with a slope defined by PL[7]. The following relevant parameters apply: XP[11]: Set the voltage threshold for the derating (from this voltage the derating starts). PL[7]: determines the derating rate in Amp/Volt (slope) NOTE: Proper default values are automatically set by the drive. It is not allowed to modify these values.
Current derating in Drum/Eagle due to temperature limits	For the above mentioned drives the maximum current of the drive is now de-rated as a function of the drive temperature (TI). This allows reducing the current gradually as the drive approach its temperature limit, thus avoiding sudden over-temperature fault. The following parameters apply to this mode, the derating is done for continuous and for peak limits separately: PL[4]: Temperature threshold to start the derating of the peak limit. CL[6] : Temperature threshold to start the derating of the continuous limit. PL[5] determines the peak limit derating rate in Amp/C° CL[7] : determines the continuous limit derating rate in Amp/C° PL[6]: temperature filter for reading the derating mechanism.



Subject	Details												
	CL[5]: minimum continuous current to be reduced for. NOTE: Proper default values are automatically set by the drive. It is not allowed to modify these values.												
Current Limit monitoring flag	The current limit state can be monitored via WI[45] command or PDO 0x2301(mapped to the EtherCAT PDO 0x1A25) as follows: <table><tr><th>Bit</th><th>Description</th></tr><tr><td>0</td><td>Use continuous current as current limit (CL[1]\PL[1])</td></tr><tr><td>1</td><td>Peak current is reduced due to high temperature</td></tr><tr><td>2</td><td>Reserved</td></tr><tr><td>3</td><td>Peak current is reduced due to high bus voltage</td></tr><tr><td>4</td><td>Continuous current is reduced due to high temperature</td></tr></table> According to the bit set, the reason for the limit can be monitored. Note that bit 0 is similar to the LC command.	Bit	Description	0	Use continuous current as current limit (CL[1] \ PL[1])	1	Peak current is reduced due to high temperature	2	Reserved	3	Peak current is reduced due to high bus voltage	4	Continuous current is reduced due to high temperature
Bit	Description												
0	Use continuous current as current limit (CL[1] \ PL[1])												
1	Peak current is reduced due to high temperature												
2	Reserved												
3	Peak current is reduced due to high bus voltage												
4	Continuous current is reduced due to high temperature												
Peak limit time increased for Drum/Eagle	For the above drives the peak limit time will be 60 seconds and not the default 3 seconds												
Twitter current derating due to voltage limits	From this version all Twitter & Bee drives are now de-rated as a function of the input bus voltage. The De-rating starts at the threshold limit defined by XP[11], and decrease from its max value, with a slope defined by PL[7]. The following relevant parameters apply: XP[11]: Set the voltage threshold for the derating (from this voltage the derating starts). PL[7]: determines the derating rate in Amp/Volt (slope) NOTE: Proper default values are automatically set by the drive. It is not allowed to modify these values.												
New Errors indication for HM\HF	When HM\HF are set, some of the parameters may conflict with each other. New enhanced error indicators are introduced for such situations that will be reported via the EC command during the initiation of the homing procedure.												
Smooth Factor increase resolution	The smooth factor denoted by SF[1] sets the smoothing of the motion in milliseconds. From this version, the smoothing can be set via SF[2] to a resolution of 250µSec as follows:												



Subject	Details
	SF[2] = 0 , 0 μSec SF[2] = 1 , 250 μSec SF[2] = 2 , 500 μSec SF[2] = 3 , 750 μSec where each value increases the smoothing of SF[1] by 250uSec. For example: if SF[1] is set to 10 and SF[2] is set to 2 than the smooth will take 10 milliseconds + 500uSec due to SF[2]. Note that with this new ability the smoothing can be as low as 250uSec by setting SF[1] = 0 and SF[2]=1.
EnDat 2.2 External Temperature Sensor Reading Support	Reading of external temperature sensor through the EnDat 2.2 communication can be configured by setting CA[60] bit 0 to 1. The temperature is reported via the TI[3] command, according to the units set by the encoder. Note that when reading the external temperature, the operating status error source from the additional information is inactive.



12.2 Bug fixes 1.1.16.00 B01

Bug	Details
NMT 0x81 while servo is enabled	NMT 0x81 soft resets the drive via CANopen networks. This command should not be sent while the motor is enabled (MO=1). In some situations, sending this command could lead to power stage failure. From this version the NMT 81 is protected and if sent while the servo is enabled an EMCY message will be sent and the command will be discarded.
Feedback Error via EC 239	When enabling the motor while there is a feedback error the EC 239 was transmitted to inform a feedback error. The information about this error was not update. This was fixed. EC 239 is now reporting "Feedback in error state or not initialized" In these situations, the drive report EMCY 0x2311.
Halls Encoder mismatch – CA[111]	The CA[111] sets the permitted offset between the halls and the encoder in electrical angle internal units (512 per 360). This feature is used to detect mismatch between these sensors and possible loss of feedback. As from this version CA[111] will be used during motor enable as well and not only after the first Halls transition, hence any mismatch higher than CA[111] +60°elect. between the Halls and the encoder while the motor is enabled will be regarded as an error. Note that during this initial phase, the mismatch limit is CA[111] +60°elect, Setting CA[111] will force a commutation search the next time that the motor is enabled.
Extended Inputs wrong report via XI[1]	XI[1] reports the state of the extended digital inputs in drives that support this feature. In drives that either support CANopen or have no EtherCAT switches, the report of the XI[1] was shifted by 8 bits to the left and actually started from 256. This was fixed. Now the report is done correctly from bit 0.
Undesired motion in mode switch	When switching from Jog position (JP) to Profile Position (PA) and modifying the SP (speed profiler), the Final speed (via FS) was set to the value of the JP . This was fixed. Now when setting JP->SP->PA the final speed is zeroed.
Recorder wrong state after LD and RS	After LD (load parameters) and RS (reset parameters) commands, the recorder status was wrongly reported as "waiting for trigger". Status is reported via RR or SR command. This was fixed, Now the status is reported as "Invalidated" – RR is -1.



Bug	Details
CANopen IP mode wrong EMCY	In Interpolated Position mode when the set point buffer was underflow a wrong EMCY message was transmitted. This was fixed.
Motor on procedure with Hot Plug in	The hot plug-in allows the user to enable the servo while the motor spins. This feature is set when the Motor Constant is set via FF[6]. In rare situations and when the FF[6] was set the following phenomena could be seen: • Stray communication watchdog: typically “heartbeat” EMCY message. • Feedback error • Motion jump after the servo enabled All the above were fixed.



Chapter 13. 01.01.15.10 B02

Component	Version
Firmware Version	01.01.15.10 30Oct2019B02
Communication Server	1.1.8.1
PAL	90
ESI	0x00010420 V11

13.1 Details

Subject	Details
Hiperface error reports	If the Hiperface sensor reports an error (i.e. CRC/Checksum) at any time, the drive status LED will become RED and the amplifier status flags, reported in bits 0:3 of the SR command will report 9.
Home Attain bit	The Home Attain bit is now reported via the Extra Status register (Object 0x2085). The bit is set when a homing process done via DS402 Homing mode was completed successfully. The bit will remain set regardless of the motion mode. The bit is reset to 0 in the following events: • Power up, CAN NMT 0x81 • RS command • New home request is received via the DS402 Control word
Dual Use Support	Based on the EU Dual use regulations and US Dual use regulations the electric frequency is limited to 599 Hz. As from this version drives with suffix "Q" that support this feature will be limited according to the regulation. Any previous firmware loaded to a "Q" drive will fail to enable the servo.
Fast Reference Torque	ID 39 sets a fast reference torque command to the drive allowing the use of current feedforward in any motion mode. The setting of the current is done via Object 0x2005 and is calculated similar to all the current\torque references as 1/1000 of rated torque.
New DRUM\ EAGLE drives	DRU & EAG 70/880 and 100/880 are now supported.



13.2 Bug fixes 1.1.15.10 B02

Bug	Details
Switching from IP mode to PP mode	In CANopen, switching from Interpolated Position Mode to Profile Position mode could sometimes cause an offset. This was fixed.
XP[1] : Overvoltage report\setting	The XP[1] allow the user to lower the overvoltage protection in order to get a more margin for this indication. The XP[1] was limited to the BV and not to the actual over voltage set by the factory. This was fixed. The XP[1] is now limited to WI[35] . For compatibility reason the value will be saturated to the WI[35] w/o an error.
Masked FoE Errors	Some Errors reported during File Over EtherCAT (FoE) were masked. This was fixed. All errors are not reported.
Max Profile Velocity units	The Object 0x607F is the maximum permitted profile velocity limit was calculated in encoder counts only and not in User Units. This was fixed.
Ignore illegal Hall bit	CA[16] bit 1 allowed the user to request the drive to ignore illegal Halls after the initiation. This allowed the user to ignore the noises that Halls may produce during run time. This bit was reset after the first servo enable preventing the protection and causing stray alarms. This was fixed.
Gantry current limit	The Gantry slave current command was not limited properly when it reached the CL/PL saturation. This is fixed. The master adjusts the command to the slave limit conditions.
Total Current Command recording	The current command recording done via the EAS recorder was recorded before the current limit resulted-in a wrong interpretation of the current state. This was fixed.
Motion jump after Stop command	When a socket reference is used the ST command caused a jump in the motion. This was fixed.
Wrong DS402 state transition	In rare situations when the fault deceleration is very low and the master sends Control word = 0 during fault reaction state, the drive reported an illegal state. This was fixed.
Serial sensor checked in Stepper motion	In stepper motion the feedback does not take any roll. The drive reported a fault if an open loop stepper was used (UM=3) while the sensor is in error state. This was fixed.
Addressing Elmo's sub index > 255	When using Objects 0x3000 and upwards to address Elmo's legacy, addressing sub-indices higher than 255 failed with an error. This was fixed. Using the 0x3000 concept as described in "Elmo Commands via Object.pdf" is now available.



Chapter 14. 01.01.15.05 B00

Component	Version
Firmware Version	01.01.15.05 01May2019B00
Communication Server	1.1.8.0
PAL	90
ESI	0x00010420 V11

14.1 Details

Subject	Details
I ² T filter in % via tPDO (rms current filter)	The peak limit filter can now be retrieved via the Application Object 0x2203 by setting object 0x2F41 bits 16-19 to 8. When the value is 100% the filter limits the current to CL[1] .
New High power Gold Twitter	G-TWI with power rates of R45/120 & R40/200 were added.
Additional speed reported via tPDO	Additional Speed can be retrieved according to the socket pointed by CA[79] . The additional speed can now be mapped to the Application Object 0x2203 by setting object 0x2F41 bits 16-19 to 7.
New Gold OBOE	OBOE 16/230 and 13/230 are introduced.
Position is not updated when CRC error in serial sensors	When a sensor CRC error is detected by the drive the position is not updated. Previously the position reported 0.
Serial sensors error & warning indications	Serial Encoder error status is reported as follows: • SR command (object 0x1002) bits 0-3 are set to 9 • EC=200 will be set if the motor enable attempt is done while sensor is in error state • Motor fault (MF) 1 is reported if an error occurred during operation state (servo enable) with adequate emergency • EE[1] will indicate the reason for the error • Drive status LED will be red Sensor Status raw data can be retrieved via object 0x2084 (OV[49] alias command). Note: During Sensor initiation the status LED will indicate Red.
Serial Sensor temperature reported via object	Temperature BiSS sensor that supports this feature is now available. TI[3] command reports the value as well as object 0x22A5 . If temperature is not available, the reported value is set to 0.



Subject	Details
	The Temperature is reported via the map able Application object 0x2203 when setting object 0x2F41 bits 16-19 to 9. (ESI modification)
Panasonic 20 & 23 bits new Error report format support	In Panasonic absolute sensors the new format of the information data field for 20 & 23 bit encoders is now supported. This includes the BLA (Battery Alarm) report. In the 17 bit models this field is 0. The BLA is reported in "Battery warning" bit 6 of Extra Status Register (0x2085).
Panasonic Sensor Status Report	The sensor status report was changed to be more convenient by reporting the same status for both Single-turn and Multi-turn feedbacks. For details refer to EE[1] Table 4 in the Gold Command Reference.
Red LED indication when sensor fails	The drive status LED is RED when sensor fault is detected. This new feature is relevant to: • All the absolute sensors faults, including sensor initialization • When analog sensor amplitude threshold is violated • When the halls are in illegal state in "Halls Only" feedback



14.2 Bug fixes 1.1.15.05 B00

Bug	Details
Current command limitation	Objects 0x6072, 0x6073, 0x60E0, 0x60E1 are now limiting the current command to the current loop. Previously only the set points were limited causing in some rare situations undesired overshoots.
Object 0x60E3 : Homing methods	Methods -3, -4 and 37 were missed from the Object 0x60E3 . This was fixed.
Homing speed limits	The Speed limit during a homing procedure was not limited to the Speed reference Limit denoted via VH[2] . This was fixed.
Emulation disabled after parameter loading	The emulation, denoted via EA[1] , was disabled after the LD command. LD loads the non-volatile parameters from the flash and activates any required function if it was activated prior to the loading. The issue was that even if the emulation was activated i.e. EA[1]=1 , than after the loading the emulation was disabled i.e. EA[1]=0 . This was fixed.
Enable servo while serial sensor in Error	The Servo Enable procedure worked even when a serial sensor reported an error. This incorrect behavior caused an undesired brake release. This was fixed. The error is now reported prior to the servo enable and the brake release is prevented.
Inverse Warning indication in BiSS	General BiSS sensor the warning bit was inversed causing a wrong indication via the Status Word warning bit. This was fixed.
Serial sensor issues after 32 bits roll-over (Halt & MT reset)	Halt after the absolute sensor rolled over 32 bits (internal count) caused an undesired gap between the feedback and the position command. This was fixed. Multi-turn reset : the actual position was not presented properly resulting in a wrong absolute position after power up cycle. This was fixed.
Setting RM=0 or EI	In some situations setting the RM=0 caused the drive to loss communication. This was fixed.
PWM emulation accuracy	In some situations the PWM emulation report was not accurate due to float rounding. This was fixed.
Tamagawa sensor warning not cleared	Tamagawa warnings were not cleared from the data report to the user. This resulted in a warning to be appeared as an error.
Halls Only wrong speed	In some situation when the a Hall changed direction at the same sampling cycle a wrong speed was calculated which



Bug	Details
	sometimes resulted in a negative speed direction. This was fixed. The speed is calculated properly.



Chapter 15. 01.01.15.00 B00

Component	Version
Firmware Version	01.01.15.00 09Oct2018B00
Communication Server	1.1.7.98
PAL	90
ESI	0x00010420 V10

15.1 Details

Subject	Details
Digital output indication when motion status 0 enhancement	The user can set the digital output to indicate that a motion reached the desired set point. This is done via the OL[] command. The indication was related to the Target Reached bit which is not valid in time based motion such as IP mode and CSP mode. From this version the digital output will also be set when the motion is in reset i.e. when consecutive set points are the same resulting zero speed command.
Object 0x100B boot version	Object 0x100B is now reporting the boot version. Previously this object was reserved.
Ignoring Gantry YAW	ER[5] is the maximum permitted tracking for Gantry Yaw controller or Stepper Closed loop controller. In some situations e.g. during tuning of the controller there might be a need to discard this error monitoring. From this version when ER[5] is equal to 0, Yaw controller or Stepper Closed Loop controller tracking error is ignored.
Sensor less observers: Using the velocity observer allows estimation of the speed for feedback purposes when no physical feedback exists	Sensor ID 37 for Brushless (Special Firmware is required): The parameters BF[1-10] are used to define the sensor properties. The commutation is also estimated. Adequate windows for the sensor-less configuration are available in the "Expert Tuning" in EAS version 2.6.
	Sensor ID 38 for DC Brush: The parameters SC[15] defines the sensor resistance and NF[11-12] are used to define the observer resolution and voltage threshold.
Halls Only speed readout	Advanced algorithm was designed which improves the halls readout significantly.
New Gold Whistle	G-WHI with power 25/100 was added.



Subject	Details																						
Profile conditioning enhanced	The profile conditioning can now also be operated in Cyclic Synchronous Position Mode (CSP). Prior to this version, the feature could be only activated on Profile position and Profile velocity modes. Note that due to this enhancement the Smooth Factor denoted by SF command might influence the trajectory of the CSP mode as well. If this is not required SF[1]=0 should be set.																						
Application Object 0x2203 enhancement	The EtherCAT application Object 0x2203 is used to retrieve process data from different signals. Refer to Ver. 01.01.13.00 B02 section 20.1 Details for a full explanation and list of prior values added. This enhancement includes extension of the “Value” options 7, 8, and 9: <table><tr><th>Value</th><th>Description</th></tr><tr><td>0 (default)</td><td>In Duet Stepper: 16 bits A2D CHA & 16 bits A2D CHB Other drives: Drive Temperature</td></tr><tr><td>1</td><td>Amplitude of the analog sensor ($\sin^2 + \cos^2$), ~2,380,000 to 2,750,000 for 1Vp-p sensor.</td></tr><tr><td>2</td><td>Speed following error in user units</td></tr><tr><td>3</td><td>Drive temperature in °C (TI[1])</td></tr><tr><td>4</td><td>Analog input 2 (AN[2])</td></tr><tr><td>5</td><td>Shunt Counter (see Note below)</td></tr><tr><td>6</td><td>VQ, the current controller output voltage</td></tr><tr><td>7</td><td>Additional Speed from socket pointed via CA[79]</td></tr><tr><td>8</td><td>Filtered RMS current in %: peak\continuance ratio for current saturation. (Ver 1.1.15.3)</td></tr><tr><td>9</td><td>Serial sensor temperature as reported from the sensor (if available, 0 if not) (Ver 1.1.15.4)</td></tr></table>	Value	Description	0 (default)	In Duet Stepper: 16 bits A2D CHA & 16 bits A2D CHB Other drives: Drive Temperature	1	Amplitude of the analog sensor ($\sin^2 + \cos^2$), ~2,380,000 to 2,750,000 for 1Vp-p sensor.	2	Speed following error in user units	3	Drive temperature in °C (TI[1])	4	Analog input 2 (AN[2])	5	Shunt Counter (see Note below)	6	VQ , the current controller output voltage	7	Additional Speed from socket pointed via CA[79]	8	Filtered RMS current in %: peak\continuance ratio for current saturation. (Ver 1.1.15.3)	9	Serial sensor temperature as reported from the sensor (if available, 0 if not) (Ver 1.1.15.4)
Value	Description																						
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2	Speed following error in user units																						
3	Drive temperature in °C (TI[1])																						
4	Analog input 2 (AN[2])																						
5	Shunt Counter (see Note below)																						
6	VQ , the current controller output voltage																						
7	Additional Speed from socket pointed via CA[79]																						
8	Filtered RMS current in %: peak\continuance ratio for current saturation. (Ver 1.1.15.3)																						
9	Serial sensor temperature as reported from the sensor (if available, 0 if not) (Ver 1.1.15.4)																						



15.2 Bug fixes 1.1.15.00 B00

Bug	Details
Sensor ID 16 not active	The analog input 1 can be used as a real time reference signal for all motion modes via sensor ID 16 (EAS Tuner). When this ID was used the reference did not work and no signal was sensed. This was fixed.
CANOpen LSS partial drive serial number reported	The drive serial number which was derived from the LSS protocol was partially reported. This resulted in some situations in an incorrect identification and a failure. This was fixed. The whole LSS message of the CANOpen is now properly reported.
Communication time out in CSV & CST	In Cyclic Synchronized Torque and Cyclic Synchronized Velocity modes, if the communication was disconnected and no new frames (set points) arrived to the drive other than the previous set points, these set points were used for endless motion. This undesired behavior is now fixed and the motion stops according to the Quick stop option code (Object 0x605A). Note that in CST the quick stop does not stop the motion but can only disable it.
Current Control modes in UM=2	When UM=2 is used (unit mode is Velocity controller) torque motion modes could not be activated. This means that the Profile Torque and Cyclic Synchronous Torque modes were not functioning properly. This was fixed. When the drive in UM=2 the user can activate the torque modes as presumed.
DV[5] : speed command report	DV[5] reported an incorrect velocity value. This was fixed. The value is now reported as User Units.
Absolute sensor 32 bits data	When using SSI, BiSS & General BiSS sensor with position resolution of over 32 bits, the position readout was incorrect. This was fixed.
Halls Only read out	When using Halls Only, on rare scenarios velocity feedback was reported as extreme values. This was fixed.
Following Error via DS402 – no fault indication	The user can select the following error monitoring between Elmo legacy via ER[3] or DS402 via Objects 0x6065 & 0x6066 . This selection is done via Object 0x2F41 (bit 3) . When the DS402 was selected the fault was not triggered as expected. This faulty behavior was limited to Interpolated Position mode of the CANopen and the Cyclic Position Mode of the EtherCAT. This was fixed. The position error fault alarm will be triggered when DS402 is selected.



Bug	Details
ST command in stepper open loop	In Stepper Open Loop ST caused a motor run-away. This was fixed.
Lead-lag filter, wrong direction	When a Lead-Lag filter is used (EAS tuner) the sign of the filter reversed causing an incorrect bandwidth and sometimes loss of loop stability. This was fixed.
Repetitive motion in User Units	The repetitive motion did not work properly when a relative set point was used and the user unit factor was not 1. This was fixed and user units can be used with repetitive motion as well. Note that the first relative motion is from the present Position Command location.
Setting capture to infinite cycles	HM[] is used for homing and capture of position. When the value is set to 32000 the capture is infinite. Previously setting the value to 16384 was also treated as infinite. This was fixed.
Relative object indexing in EtherCAT	For Object 0x3xxx the use of indices higher than 255 is done via the offset. Using the offset was not permitted and prevented the ability to address indices higher than 255. This was fixed.
Limit the speed in UM=3	The speed in stepper open loop was wrongly limited by VH[2] . This was fixed.
Gantry Yaw error – wrong calculation	When the sensor 32 bits was rolled over, the Gantry Yaw controller error was not calculated properly . This was fixed. In situations where the sensor 32 bits is rolled over, a jump of the motion is now prevented. Note that the max permitted error limit is defined via ER[5] .
Wrong DS402 status word identification	In some situations, the status word indication during Current Identification caused a report of motor off while the motor was on. This was fixed. The wrong indication happened only during the identification done via the EAS.



Chapter 16. 01.01.14.08 B00

Component	Version
Firmware Version	01.01.14.08 17Dec2017B00
Communication Server	1.1.7.97
PAL	90
ESI	0x00010420 V07

16.1 Details

Subject	Details
Field Weakening	For situations where a high velocity is required and the voltage is limited, employ a field weakening by controlling the reactive current reference to reduce the voltage saturation due to Back EMF. FF[8] is the reference for the reactive current in elec.deg/cnt/sec. (default 0) CA[114] Speed threshold from which the FF[8] starts to take effect. Please note that the FF[8] is a new variable that will result in a parameter reset after the firmware is updated.
Phase Advance is enhanced	Phase advance was defined by a single compensation using FF[4], where positive values compensated both for current & voltage and negative values compensated for voltage only. From this version, FF[4] compensates for the delay between the sensor and the current and an additional compensation between the current and the voltage was added and is defined by FF[7]. The units of FF[4] and FF[7] are multiple of TS/2 (half of the sample time).
Object 0x603F Error Code mapped	The Object 0x603F reports the last error code and can now be mapped to PDO. In EtherCAT PDO 0x1A26 contain this object. Note that during motor enabling the 0x1A26 will be cleared.
Object 0x201B: Read the Current RMS filter	The Object 0x201B retrieved the filtered RMS current which is used for the I2T filter. This allows the host to monitor the state of the current and if required to take adequate measurements prior to current saturation. Available via the CANopen network only.
New High power Gold Twitter	G-TWI with power rates of R45/120 & R40/200 were added.



Subject	Details
Profile Conditioning	The Profile Conditioning method modifies the motion profile and as a result minimizes the vibrations in a moving system. NF[8] enables the relevant conditioning modes. NF[9] and NF[10] are used for the parameters of the selected mode. Using Profile Conditioning shortens the stabilization time and improves the profile performance.



16.2 Bug fixes 1.1.14.08 B00

Bug	Details
Current command limitation	Objects 0x6072, 0x6073, 0x60E0, 0x60E1 are now limiting the current command to the current loop. Previously only the set points were limited causing in some rare situations undesired overshoots.
Object 0x60E3 : Homing methods	Methods -3, -4 and 37 were missed from the Object 0x60E3 . This was fixed.
Homing speed limits	The Speed limit during a homing procedure was not limited to the Speed reference Limit denoted via VH[2] . This was fixed.
Emulation disabled after parameter loading	The emulation, denoted via EA[1] , was disabled after the LD command. LD loads the non-volatile parameters from the flash and activates any required function if it was activated prior to the loading. The issue was that even if the emulation was activated i.e. EA[1]=1 , than after the loading the emulation was disabled i.e. EA[1]=0 . This was fixed.



Chapter 17. 01.01.14.05 B02

Component	Version
Firmware Version	01.01.14.05 22Oct2017B02
Communication Server	1.1.7.96
PAL	90
ESI	0x00010420 V06

17.1 Details

Subject	Details
New High power Gold Twitter	G-TWI with power rates of 80/80 & R160/80 & R140/100 were added.
DS402 Homing Method 37	Homing method 37 allows user to set the Homing Offset Positing (0x607C) as the present position.
Home Attained bit indicates that home was already done	From this version bit 12 in the status word (0x6041) can be used as an indication if a homing was attained without performing the homing procedure. This indication is relevant if bit 4 in Object 0x2F41 was set to '1'. In this case when Home Mode is selected via object 0x6060.0 , bit 12 is set to '1' if the homing was successfully performed since the last power on cycle. The Home attain bit is cleared is case of feedback error.

17.2 Bug fixes 1.1.14.05 B02

Bug	Details
Profile Speed affects the Homing mode speed	Setting the profile velocity Object 0x6081 to 0 undesirably caused the homing mode to stop. This was fixed. The homing mode speed is not affected by the profile velocity object.
Fault warning from absolute sensor	Randomly, when absolute sensors are used the Warning bit of the DS402 Status Word (0x6041) was unintentionally set to 1. The cause of this was due to the incorrect initiation of the warning during power up. This was fixed.
Wrong parameter CRC calculation after reset	Due to a wrong default value of the emulation parameters EA[13] & EA[14] , the parameter CRC indicated via OV[52] or Object 0x2060 .



Chapter 18. 01.01.14.01 B00

Component	Version
Firmware Version	01.01.14.01 20Aug2017B00
Communication Server	1.1.7.95
PAL	90
ESI	0x00010420 V06

18.1 Details

Subject	Details
CANopen Applications	This version was created to provide a firmware version only recommended for CANopen applications. Consult Elmo Support prior to using it in EtherCAT based applications.

18.2 Bug fixes 1.1.14.01 B00

Bug	Details
Delay of ~30secs during power up.	Sometimes after power up, a delay of about 30 seconds suspends communication to the drive. This was fixed. NOTE: when the delay occurs, the Safe Torque Off (STO) diagnostics reports an error that prevents any attempt to enable the motor.
No deceleration & acceleration limits in PTP mode	In profile Position mode (PTP) the deceleration (DC) and acceleration (AC) were not limited by the Stop Deceleration value (SD) as expected. This was fixed.



Chapter 19. 01.01.14.00 B01

Component	Version
Firmware Version	01.01.14.00 05June2017B01
Communication Server	1.1.7.94
PAL	90
ESI	0x00010420 V06

19.1 Details

Subject	Details
Safe Torque Off (STO) version 14 certified.	The STO diagnostic tests were modified according to IEC61800-5-2 with one of the tests removed as superfluous. This was approved by TUV SUD report number: EP85169C-A1 from 09/05/2017. STO version is retrieved via VS command: 1.0.14.0
Panasonic serial sensor new resolutions	Panasonic serial sensor with 20 & 23 bits resolution is now supported.

19.2 Bug fixes 1.1.14.00 B01

Bug	Details
UI[5] to UI[8] wrong default value	The UI[5] to UI[8] indexed default values (after RS command) were not set to 0 as expected. In addition, these parameters did not set correctly when negative values were introduced. This was fixed.
OV[] and OF[2] to OF[6]	Some of OV[] parameters and OF[2] to OF[6] did not updated when a legal value was set to them. This could cause a wrong behavior of the alias objects of these variables. This was fixed.
RS232 disconnected after RS command	The RS232 communication was disconnected after the RS command was invoked. This was fixed.
Stepper commutation did not abort	Sometimes when stepper commutation was activated during the motor enable process, if the commutation procedure failed, the motor remained in servo. This was fixed. The motor is now disabled with the MF 2 : "Commutation sequence Failed" message.
General BiSS sensor failure	In some situations the General BiSS sensor caused the drive to fail. This was fixed.



Chapter 20. 01.01.13.00 B02

Component	Version
Firmware Version	01.01.13.00 20Apr2017B02
Communication Server	1.1.7.94
PAL	90
ESI	0x00010420 V06

20.1 Details

Subject	Details																
Position tracking error fault via DS402 Position Following window	Starting from this version, the Position Following Error can be determined via Objects 0x6065 & 0x6066 (Following Error Window & Following Error time out). By setting bit 3 of Object 0x2F41 to '1', the position following error fault (MF = 128) will be determined by these two Objects instead of the legacy ER[3] command. This means that when the Position controller following error is greater than Object 0x6065 for a longer period than 0x6066 , the drive goes into Fault state with MF and the relevant EMCY message. If bit 3 of 0x2F41 is set to '0' (default) the position following error is determined by the ER[3] value.																
Application Object 0x2203 enhancement	The EtherCAT application Object 0x2203 is used to retrieve process data from different signals. Signals are selected via Object 0x2F41 bits 16 to 19. From this version VQ (active output voltage of the Current Controller) can also be recorded. The following summarizes the optional variables that can be mapped to 0x2203 – the "Value" is Object 0x2F41 bits 16-19 : <table><tr><th>Value</th><th>Description</th></tr><tr><td>0 (default)</td><td>In Duet Stepper: 16 bits A2D CHA & 16 bits A2D CHB Other drives: Drive Temperature</td></tr><tr><td>1</td><td>Amplitude of the analog sensor ($\sin^2 + \cos^2$), ~2,380,000 to 2,750,000 for 1Vp-p sensor.</td></tr><tr><td>2</td><td>Speed following error in user units</td></tr><tr><td>3</td><td>Drive temperature in °C (TI[1])</td></tr><tr><td>4</td><td>Analog input 2 (AN[2])</td></tr><tr><td>5</td><td>Shunt Counter (see Note below)</td></tr><tr><td>6</td><td>VQ, the current controller output voltage</td></tr></table>	Value	Description	0 (default)	In Duet Stepper: 16 bits A2D CHA & 16 bits A2D CHB Other drives: Drive Temperature	1	Amplitude of the analog sensor ($\sin^2 + \cos^2$), ~2,380,000 to 2,750,000 for 1Vp-p sensor.	2	Speed following error in user units	3	Drive temperature in °C (TI[1])	4	Analog input 2 (AN[2])	5	Shunt Counter (see Note below)	6	VQ , the current controller output voltage
Value	Description																
0 (default)	In Duet Stepper: 16 bits A2D CHA & 16 bits A2D CHB Other drives: Drive Temperature																
1	Amplitude of the analog sensor ($\sin^2 + \cos^2$), ~2,380,000 to 2,750,000 for 1Vp-p sensor.																
2	Speed following error in user units																
3	Drive temperature in °C (TI[1])																
4	Analog input 2 (AN[2])																
5	Shunt Counter (see Note below)																
6	VQ , the current controller output voltage																



Subject	Details	
	7	Additional Speed from socket pointed via CA[79] (See Chapter 3 Ver 1.1.15.0)
	8	Filtered RMS current in %: peak\continuance ratio for current saturation. (Ver 1.1.15.3)
	9	Serial sensor temperature as reported from the sensor (if available, 0 if not) (Ver 1.1.15.4)
	Note: Shunt Counter : The shunt counter used in Bassoon and Oboe drives can now be retrieved via the above object. The counter is 32 bits and counts the time when the shunt is active and dissipating regeneration power. When the counter reaches 2,500,000 the shunt is disconnected.	
EtherCAT scan is discarded during servo on	The EtherCAT master scans the network to identify EtherCAT devices. This scan happens if a device was connected to the network. During the scan no EtherCAT message is processed. This means that the process data fails to arrive in time and as a result any motion which is time dependent (such as Cyclic Synchronous Position) must use an extrapolation in order to maintain the motion. From this version, in order to prevent the extrapolation in such scenarios, the scan request from the host will be discarded when the motion is enabled. The host should expect a timeout in such situations.	
Recording the Digital Halls	The Digital Hall recorded signals are now available via the EAS recorder for any user. The “Hall Sensor” signal indicates the halls pattern.	
Warning Bit of general BiSS is now reported in 0x2085	If CA[60] bit 10 is set, the warning of a BiSS sensor will be reported via bit 6 of the map-able Object 0x2085 . This feature is available for BiSS sensors where the Warning Bit arrives immediately after the sensor Error Bit.	
IA Command – not valid	From this version, the IA command is no longer active. For compatibility reasons, the command will not return an error when set or requested.	



20.2 Bug fixes 1.1.13.00 B02

Bug	Details
SDO message is discarded by EMCY message via EtherCAT channel	In rare situations, when an EMCY message and a reply to an SDO message are required at the same time, the SDO message was discarded. The situation caused a timeout at the host. This was corrected.
Following Error was not indicated in the DS402 Interpolated Position Mode	In position motion modes bit 13 of Object 0x6041 (Status Word) indicates the following error. When the drive was in Interpolated Position Mode (Object 0x6060 = 7 in CANopen) the bit was ignored. This was corrected.
Setting OV[37] was overwritten	When setting the Home Speed via Object OV[37] , the actual value was not reported correctly. This was corrected. Note that the same issue can happen when other OV[] variables are read.
EtherCAT ID switches after firmware downloading via FoE	In rare situations, the EtherCAT identification switches were reported 0 after firmware downloading via FoE. A power up cycle was required to recover the identification number. This was corrected. From this version the EtherCAT ID is reported properly after firmware update via FoE. Note that PAL 90 is required.



Chapter 21. 01.01.12.02 B00

Component	Version
Firmware Version	01.01.12.02 13Feb2017B00
Communication Server	1.1.7.90
PAL	89
ESI	0x00010420 V06

21.1 Details

Subject	Details
Friction compensation in Synchronous Cyclic Position (CSP) mode	Friction compensation is now allowed also in CSP mode and when Object 0x60B1 is mapped. Friction compensation is set via NF[1] and NF[2] command (Amps) for positive and negative directions respectively.

21.2 Bug fixes 1.1.12.02 B00

Bug	Details
Gantry Homing was not accurate during high speed	When Gantry homing was performed at high speed, the actual position (PX) value and Object 0x6064 (PU) were sometimes unexpectedly different. This was corrected.
When XP[2]>2 , analog inputs readout may not be properly synchronized	The XP[2] command allows the user to modify the frequency of the control PWM. When XP[2] was set to a value higher than 2 the synchronization of the analog inputs may not have been correct. This was corrected. Note that by default XP[2] equals or is lower than 2.
Undesirable Reset of the absolute position in Tamagawa sensor	Sometimes when a Tamagawa sensor is used and the user loads the parameters via the LD command or resets the parameters via the RS command, the absolute position was reset. This was corrected.



Chapter 22. 01.01.12.00 B01

Component	Version
Firmware Version	01.01.12.00 14Nov2016B01
Communication Server	1.1.7.90
PAL	89
ESI	0x00010420 V06

22.1 Details

Subject	Details
Improving velocity readout in Sine\Cosine sensors	When an analog sensor is in high frequency there is an option to read the position and the speed of the quadrature signals rather than the analog signal. This feature overcomes a possible distortion due to the analog filters. CA[112] sets the threshold speed from which the analog signals are ignored and the speed and position are updated from the quad signals. CA[112] is the analog frequency in Hz. If the value is 0 (default) the function is ignored and the speed is given from the analog signals. The hysteresis threshold for resuming to analog readout is 80%.

22.2 Bug fixes 1.1.12.00 B01

Bug	Details



Chapter 23. 01.01.12.00B00 (same as 01.01.11.16 B00)

Component	Version
Firmware Version	01.01.12.00 14Nov2016B00
Communication Server	1.1.7.90
PAL	89
ESI	0x00010420 V06

23.1 Details

Subject	Details
Target Reached in Stepper Closed loop (UM=6)	From this version, when the drive is in Stepper Closed Loop, the Target Reached bit (0x6041 , bit 10) is available. The indication of the motion status is available via the MS command as well. Before this version the target reached indication function was ignored in stepper closed loop.
Analog input 2 readout via EtherCAT process data	From this version, the analog input 2 (AN[2]) can be mapped to Object 0x2203 and read via EtherCAT Sync Manager 2 (PDO 0x1A24). By setting bits 16-19 of Object 0x2F41 to 4, the analog input is mapped. The value of the object is the same as the AN[2] command.
Emit() function added	The legacy emit() function is used to transmit an EMCY message from the User Program to the CAN network. This function is now implemented in the Gold Family as well. The emit(data) results in a transmission of EMCY message: 0xFF01 0x81 and the data.
Lead\Lag first order filter enhancements	The Lead\Lag first order filter can be set via the KV[] command. From this version, the phase can be set to +/-85 degrees, previously +/-60 degree, and the frequency to 0.2Hz, previously 1Hz.
Plant identification improvement	The frequency used for the identification of the plant during the tuning (SE[3] command) is now set to 6667 HZ allowing identification at higher frequencies. Previously the highest frequency was limited to 5000Hz.



23.2 Bug fixes 1.1.12.00 B00

Bug	Details
Sync Manager 3 processing failed in SafeOP state	In SafeOP state, SM3 was not processed properly causing the message from the drive to be discarded. This was corrected.
PDO 0x1A03 – not aligned	PDO 0x1A03 was not aligned, causing an incorrect readout value of any object which followed it. This was corrected.
SDO segmented process failed	In some situations, the use of commands handling objects directly (i.e. OF[], OV[], CW & FC) caused an on-going SDO segmented process to fail with a dead-lock situation. This was corrected. The use of these commands is now permitted without causing the mentioned failure.
Object 0x3xxx wrong handling of the index	No indication when the sub-index of Object 0x3xxx is out of permitted range. From this version, the sub-index of Object 0x3xxx is handled properly.
Homing after PA\PR command	When the PR command is followed by the PA command, using Homing caused an undesirable move, compensating the PR value. After the target is reached, this undesired motion is now inhibited. Even setting a homing after the target was reached, does not cause any motion. However, setting a homing during the motion will cause a recalculation of the target which results in a motion to the new position as a result of the homing command.
Undesirable motion when setting PA command to a limit switch while in ECAM mode	When the axis encountered a FLS\RLS limit while the ECAM is active and the PA command is used, an undesirable motion occurs. This was corrected. Now if the ECAM command with combination of the PTP command is beyond the limit switch position, then the motion towards the switch is inhibited.
Jog command (JV) violated the limit switch (FLS\RLS) when FF[1] is used	If the acceleration feedforward command is used (FF[1]! = 0) and a jog motion encounters a limit switch (FLS\RLS), the motion did not stop as expected. This was corrected. The limits are not violated in such situations.
Position Limit in stepper closed loop did not work properly	In Stepper closed loop, when a software position limit was reached, the motor could not be moved out of the limit. This was corrected.
Mandolin temperature report	In the G-Mandolin drive the temperature was reported incorrectly with an offset of ~3°C. This was corrected.



Chapter 24. 01.01.11.11 B00

Component	Version
Firmware Version	01.01.11.11 12Sep2016B00
Communication Server	1.1.7.89
PAL	89
ESI	0x00010420 V06

24.1 Details

Subject	Details
Download parameter file in PreOP state	The parameters file can be downloaded via FoE. Before the modification the FoE download was permitted only in Boot Strap state. From this version the file can be downloaded during PreOP state as well. Note that firmware download via FoE is allowed only in Boot Strap state,
Bassoon\Oboe shunt ability	The regeneration power dissipation of the G-BAS & G-OBO was further improved.

24.2 Bug fixes 1.1.11.11 B00

Bug	Details
Trigger on Begin	When the BG command was requested prior to the pre-trigger delay time, the recorder failed to start. This was fixed. The recorder starts when the pre-trigger elapses.
Yaskawa sensor CRC	In some situations, when Yaskawa sensor is used, an incorrect CRC error was detected. This was fixed. The modification requires the use of PAL Version 89 (VP).
ESI fixes	The following was fixed: - Object 0x1003 was missed - Object 0x60FD type change from DINT to UDINT



Chapter 25. 01.01.11.09 B00

Component	Version
Firmware Version	01.01.11.09 12Jul2016B00
Communication Server	1.1.7.87
PAL	88
ESI	0x00010420 V04

25.1 Details

Subject	Details

25.2 Bug fixes 1.1.11.09 B00

Bug	Details
Analog input 1 offset	The analog input 1 offset was not calculated properly.This was fixed.



Chapter 26. 01.01.11.08 B00

Component	Version
Firmware Version	01.01.11.08 29Jun2016B00
Communication Server	1.1.7.87
PAL	88
ESI	0x00010420 V04

26.1 Details

Subject	Details
Trapezoidal commutation	Setting CA[28]=8 will allow commutation in real 6-step trapeze mode. This improves the performance of the servo when the motor windings are a real trapezoidal. This mode can be used with DC-brushless where Halls Only feedback is used. The main improvement is to maintain a smooth torque when the motor switches direction.
Ignore digital Halls fault after commutation was found	Setting CA[16] bit 1 to '1' indicates to the drive, to ignore the Halls state, when the commutation is located in a Halls+Encoder sensor. When there is an illegal Hall state, bit 10 of object 0x2085 (indicated via the setting of OV[61]). The bit is cleared when the next motor enabled (e.g. MO=1) is commanded.
Ignore digital Halls mismatch fault	CA[111] sets the mismatch field angle between the Halls and the Encoder in field angle units (512 is 360 electrical angles). Setting CA[111] to 0 disables this function. If the difference between the field angle calculated by the Halls, and that calculated by the encoder is higher than CA[111], the drive disables the servo with MF=4. This fault can be ignored by setting CA[16] bit 2 to '1'. If the "ignore" bit is set and a fault occurred, Object 0x2085 bit 11 is set to indicate a warning. The warning is also reflected via OV[61] bit 11.
Object 0x201B retrieves the current RMS	The current RMS is calculated for the CL\PL protection filter. This value can now be retrieved via the map able object 0x201B. The object is map able via CAN only.
Gantry applications, CA[60] bit 7: Allows separation of the analog current offset for master and slave	For a Gantry application during identification, CA[26] points to the socket used by the master to produce the Yaw torque also for the slave. If the user wishes to use CA[26] for current offset in the master, setting bit 7 of CA[60] will prevent the master



Subject	Details
	from sending the Yaw torque. This means that the CA[26] socket of the master will now serve as a reference to the master only.
Home supported in closed loop Stepper Mode	In UM=6 and closed loop Homing modes are now valid.
Mandolin drive	The Mandolin drive was added to the Gold family.
Interpolated Position (IP mode) synchronization enhancement.	The CANopen IP mode requires a SYNC message to synchronize the motion between several drives. SYNC is a CAN message produced by the host, so that the drives adjust their internal timers. From this version, the drive tolerance for SYNC message jitters and drifts is improved. Previously, the drive reported underflow due to the SYNC message drifts. Note that the minimum SYNC interval in IP mode is 1mSec. The SYNC interval determines the motion interpolation period via object 0x60C2. The drive does not check this period The host must guarantee the interpolation time to avoid buffer underflow.
Reading QUAD sensor from PAL	Sensor ID 36 allows reading of a quad sensor based on the CPLD hardware interfaces. This feature allows very high resolution timing synchronization (<20 nanosecs) of quad signals lock, which is useful in some cases for high speed quad signals (> Few MHz Pulse rates). In this mode, capture and Homing are limited.
Prevent movement when Hard Stop is active	When Hard Stop is active (IL[] command :Soft & Auxiliary stop), the drive will not perform commutation search. Any attempt to enable the servo (e.g. MO=1) when the commutation is not known, receives the reply EC 201: “Commutation Sequence Failed”. The user should reset the input prior to enabling the motor. Note that If the commutation is known, or if the sensor allows absolute commutation (e.g. Halls sensors) the servo can be enabled even when Hard Stop is active.
External 16 bits A2D sensor	In the Duet Stepper, the drive can now read data from two 16 bits A2D channels for general purpose. The A2D channel A & channel B values are reported via WI[70] & WI[71] respectively. These values are also reported via Object 0x2203 where the high 16 bits are the readout from channel A and the low 16 bits are from channel B of the A2D.



Subject	Details
Interpolation time threshold between Halls transition	In Halls only sensor, CA[40] is the threshold interpolation time between digital halls transitions in mSec. If the time between transitions is higher than CA[40] , then the speed interpolation is disabled. Setting CA[40] to 0 means that the default interpolation is 2000 mSec.
Enhanced stop of motion when drive in saturation	When the drive is in voltage saturation, meaning that the bus voltage is so low, it prevents the motor from following the required speed command, and a speed tracking error is developed between the feedback and the speed command. When the motion is stopped (with the ST or Halt command) in these situations, the feedback continues to move until the tracking error fades. This undesired movement of the motor may last for a relatively long period after the speed command was actually set to 0. From version 1.1.11.4, the tracking error in such situations, will be closed immediately, allowing the drive to stop the motor at the desired deceleration regardless of the tracking error caused by the voltage saturation.
Interpolated Position active state & underflow modification	The CANopen IP mode starts when the host sends the SYNC messages and sets bit 4 of object 0x6040. If the host fails to send the first set-point (object 0x60C1) prior to the 0x6040 setting, the drive assumes a failure and informs that there is an underflow by resetting the interpolation active flag (object 0x6041 bit 12). From version 1.1.11.6, the drive will not start the underflow counter before the first set-point arrives even if bit 4 of object 0x6040 (IP active) is set. This enhances the tolerance of the drive for different timings of the CANopen host.
Setting a new set-point in Profile Position mode while the drive is in Halt	When the host sends a Halt command via object 0x6040 (bit 8), the drive halts the motion according to the definition in Halt Option Code (0x605D). If during this time, the drive is halted, a new set point is issued by the host, so that when the Halt is “released” by the host, the new motion trajectory will be according to the new set-point. In previous version the new trajectory requested during Halt was discarded.
Twitter over temperature	The Gold Twitter over temperature is 87°C. Previously it was 85°C.
Home to block + Index Sub modes -3 & -4	New homing modes -3 & -4 are introduced in this version. Homing mode -3 means that the motor moves in the negative direction with limited torque and defined threshold velocity until blocked. When the motor is blocked and the torque



Subject	Details
	reaches the torque limit, if the speed is lower than the threshold speed for a threshold speed time, then the motor moves in the positive direction where the torque limit which was used before the homing is restored, until the Index is detected and the home is attained. The -4 mode is the same except that the motor moves to the positive direction until it is blocked and then to the negative direction until it detects the Index pulse.
FLS & RLS during homing modes	The reverse and forward limit switches were disabled during homing mode. Starting from this version during the DS402 Homing Mode, the FLS & RLS are monitored. If the drive detects one of the mentioned limits and this limit is not part of the homing procedure, then the motor stops and Home Error bit is set (Object 0x6041 bit 13). The limit can be monitored via 0x60FD.
Improving the current identification frequency range	The frequency which is used for frequency identification is now improved. Previously the lowest frequency was 1Hz. From this version onwards, it can be set as low as 0.01Hz.
Position Software Capture of digital absolute sensors and analog sine\cosine sensors	Position capture is used for touch probe and homing functions. The capture is now permitted in all absolute sensors as well as in analog sine/cosine sensors. The supported absolute sensors are: BiSS, Panasonic, EnDAT, Tamagawa, SSI, Kawasaki & Sanyo Denki. When an absolute sensor is used for capture, the accuracy of the captured position depends on the velocity according to: Feedback velocity x TS For analog sine\cosine sensors the user can select between hardware and software capture. Hardware capture is very accurate but with low resolution, whereas the software position captures is with high resolution but low accuracy. The analog sensors which support the software capture are Hiperface, Sine\Cosine and Gurley sensor. Software capture is selected by setting bit 8 in CA[60] (CA[60] = 512). Note that the user is required to reset the sensor via CA[41] to CA[44]. When selecting hardware capture (default, compatible to previous version) the position is captured by quadrature transition of the signal. Alternatively, when selecting software



Subject	Details
	capture, the resolution is according to the analog resolution but the accuracy is limited to: Feedback velocity x TS Note that Halls sensor and Resolver cannot be captured by software. The capture is performed via the Index or Strobe inputs of Port A. Any digital input can be routed to the Index or Strobe via GI[3] or GI[4] respectively.
Temperature mapped via EtherCAT	The drive temperature can now be monitored via Object 0x2203. Object 0x2203 is a general object pointer that can retrieve different data. Object 0x2F41 bits 16 to 19 are used to select which data can be retrieved. Setting these bits to 3 means that the 0x2203 retrieves the drive temperature (alias to 0x22A3 sub index 1). For EtherCAT channel, 0x2203 is mappable via tPDO 0x1A24.
Shunt logic improved	Shunt logic software protects the drive's shunt. The shunt logic is available in G-BAS & G-OBO. The handling of the shunt logic was improved allowing a higher dissipation of the shunt power.
Downloading PAL via CAN	From this version it is permitted to download the PAL via a CAN interface. Previously, PAL downloading was limited. Note that the user should confirm the PAL version is suitable to the firmware version, meaning that the download via CAN is performed from the same FW file (*.gabs file).



26.2 Bug fixes 1.1.11.08 B00

Bug	Details
ESI modifications	Minor adjustments in data type definition.
Jump in position when stepper\binary commutation is done	Sometimes after a stepper commutation, the position jumped. This was fixed.
Possible position jump in non-periodical ECAM	In a non-Periodical ECAM (EM[1]=1) where a non-constant master gap (EM[2] bit 11 set to 1), if the ECAM starting point was non-zero, a position jump occurred out of the ECAM. This was fixed.
SR bit 21: Shunt indication	Shunt indication was reversed and the bit was set when shunt was inactive. This was fixed.
G-TUBA internal fan constantly active	When downloading the previous version of firmware to G-TUBA the internal fan continues to operate although switched OFF. This was fixed. The G-TUBA fan only works when the temperature threshold is reached.
Wrong deceleration during Position limit reaction	When the drive reaches the position limit it automatically decelerate the motion using the SD command. In some situations the end of this deceleration was incorrectly managed, especially at the last interval of the motion causing an actual deceleration which is higher than the one defined via SD . This was fixed.
Speed 0 indication in motor off	Bit 23 in SR command reports when the speed is 0. This bit reported incorrectly when the servo was off i.e. SO=0 . This was fixed.
Fail to engage brake in torque modes	When the drive is in Profile Torque or Cyclic Synchronous Torque modes and a brake function is required via BP[] , the brake was not engaged when the drive was disabled. This was fixed.
CANopen: RTR not supported	RTR allows a remote transmission mechanism of the CANopen. The Gold drive does not support the RTR, which is informed via Object 0x1800 to 0x1803 sub index 1 bit 30. Note that for compatibility reasons, if the host sets bit 30 to 0, the request for RTR will be ignored, without error. Drive always reports 1 in this bit.
User program: wrong code size	The allowed Code size for the user program is 30000 chars. Previously it was set to 15000. This was fixed.



Bug	Details
Software position limits during homing	During homing procedure (DS402 Operation Mode 6) the Position software limits (VH[3] \VL[3]) blocked the motion. Starting from version 01.01.11.02 these limits are ignored until the home is attained.
Object 0x0002 map able	The 0x0002 object is a byte object that allows performance of a virtual CANopen PDO mapping for padding. Previously, the object could not be mapped to both Read & Write PDOs. Now it is permitted. Note that the object is not valid for EtherCAT, which uses object 0x0000 for padding.
Interpolated Position (IP mode) with velocity reference	When using IP mode sub-mode -1 where the velocity of the next set-point is handled by the host, the velocity was not handled correctly causing jitters in the motion. This was fixed.
Motion jumps in Profile Position Halt & Start transition	In Profile Position Mode, if the host sends Halt and New Set-Point (bit 4 in 0x6040) repeatedly in quick succession, the motion jumped. This was fixed.
Hiperface sensors not working after LD	LD loads the parameters from the flash memory. The LD is automatically called during power up. If the host issued LD after the power up, the Hiperface sensor position readout was not updated. This was fixed.
Changes in the EtherCAT ESI file	The following objects were fixed in the ESI file: 0x60E3 changed type to SINT 0x60c2.2 changed type to SINT 0x608F, 0x6091, 0x6092, 0x6099 changed type to UDINT
Recorder trigger on Begin	The BG recorder trigger started the recording prior to the trigger delay timing. This was fixed.
Interlock polarity modified	The Interlock bit is reset to '0' in object 0x60FD when STO is active and the drive is in safe state. Prior to this version the logic of the Interlock bit was wrongly inversed.
A cycle delay when tPDO event is triggered on digital inputs,	When a tPDO is set for a Asynchronous event with a Digital Input trigger, the transmitted data of the digital input was from the previous cycle. As a result, the transmitted digital input data were wrong. This was fixed. The integrity of the digital inputs which are transmitted via tPDO is now corrected. Note that this issue did not occur when tPDOs were triggered by the SYNC message.



Bug	Details
Quick Stop during Homing procedure	When the drive is in Homing Mode Quick Stop state, a Home Error indication (bit 13 object 0x6041) occurs. In order to resume motion, bit 4 of object 0x6040 must be toggled from 0 to 1.
Homing method modification only when home is not active	In previous versions, setting a new homing method while the previous method did not complete resulted in uncertain situation. From this version, switching method, while the previous method is operational is permitted only after setting bit 4 in 0x6040 to 0.
Nikon sensor initiation	Sometimes the Nikon sensor started unexpectedly with a negative value. This is due to error reset after a battery change. This was fixed.



Chapter 27. 01.01.11.00 B00

Component	Version
Firmware Version	01.01.11.00 19Nov2015B00
Communication Server	1.1.7.85
PAL	87
ESI	0x00010420 V03

27.1 Details

Subject	Details
Prevent movement when Hard Stop is active	When Hard Stop is active (IL] command :Soft & Auxiliary stop), if during the motor enable the commutation is not known, the drive discards the enable request with EC 201: <i>"Commutation Sequence Failed"</i>. The user is required to reset the input prior to enabling the motor. Note that If the commutation is known, (no commutation search is required) or if the sensor allows absolute commutation (e.g. Halls sensors) the servo can be enabled even when Hard Stop is active.
ECAT Complete Access removed from ESI	Complete Access allows setting or obtaining of arrays and records in a single communication packet. In the Gold drives, this feature is permitted for a limited set of objects mainly Startup and mapping objects. As the Complete Access option is now removed from the ESI (0x00010420), a new EtherCAT configuration can address all objects in a similar fashion. For compatibility reason addressing the limited objects as described above with Complete Access is still permitted.
EtherCAT PDO Padding object	Padding is required during PDO mapping to maintain even octet's memory boundaries. Previously object 0x0002 was used to align 8 bit objects to 16 bits. According to ETG 2000, the padding object for this purpose is 0x00000008, which adds padding of 8 bits. Therefore, the 0x0002 object is replaced with the 0x00000008 object. AS a result of this padding object change, the following PDOs are affected: • Rx (Inputs): 0x1605, 0x160B, 0x161E (SM3) • Tx (Outputs): 0x1A02, 0x1A03, 0x1A04, 0x1A0B, (SM2) The ESI <i>Elmo ECAT 00010420 V03</i> supports this option.



Subject	Details
	Note that for compatibility reasons object 0x0002 is still valid via the on-line dictionary.
Speed range enhancements	Speed parameters can now be set to 0. Object 0x607F determines the Max Speed via DS-402, which can be set to 0. This means that the speed command is saturated to 0 for all motion modes, when the max. speed is 0. Object 0x6099 (sub index 1 & 2) determines the speed used when DS-402 Homing mode is activated. The object can now be set to 0. The SP is the maximum speed of the Profile Position. From this version, setting between SP[1] to SP[4] to 0, is permitted. If the value of SP is set to 0, the next BG command results in a Stop, similar to the ST command. Object 0x6081 is similar to SP, and sets the Profile Velocity. Setting this object to 0 is permitted. It results in a 'Halt' command when the <i>New Set-Point</i> bit (bit 4) in the Control Word is next set.
DS-402 Homing re-activation enhancement	When a DS-402 homing procedure is interrupted via the control word (0x6040) by setting bit 4 to 0, the homing sequence is considered as "failed" and a Homing Error bit is set to 1. Resetting bit 4 back to 1 initiates the homing procedure again. Previously a re-initialization of the Home Mode via 0x6060 was required.



27.2 Bug fixes 1.1.11.00 B00

Bug	Details
EMCY via EtherCAT, missed bytes	When emergency message was transmitted via EtherCAT, only the first 4 bytes were transmitted properly. This was fixed. Now all 8 bytes are transmitted.
Stop On Fault during Torque motion modes	In Torque modes (Profiled torque and Cyclic Synchronous Torque), when Fault Option Code (0x605E) is <i>Slow Down</i> before disabling the servo, in rare situations the servo was not disabled. This was fixed. Now if Torque modes are operational the servo is disabled immediately.
Position drift when Gantry is homed	In Gantry systems, the reported position after a home is attained, was offset from the actual position. If multiple homing procedures were performed, the absolute position drifted. This was fixed.
Torque increased when a higher CL[1] value is set.	When torque is applied to the motor via TC command and the CL[1] (Continuance Limit) was increased. This actual torque (IQ) was unintentionally increased regardless of the state of the torque saturation. This was fixed. Increasing the CL[1] does not affect the torque that was applied to the motor.
Object 0x20B0 sub 9 incorrect report	Object 0x20B0.9 reports the socket which is used for the Additional Position report (CA[79]). Previously, the object reported the additional position instead of the socket number. This was fixed.
XP[3] added to parameter file	XP[3] sets the filter of the CANopen SYNC message arriving from the host. The value was not uploaded by EASII as part of the Parameter file. This was fixed.
EA[4] indication of the Velocity socket for Emulation	EA[4] indicates the socket used for Velocity PWM emulation. When set to 0, the drive automatically routes the actual velocity socket (defined by CA[46]) to the PWM emulation. Previously, setting EA[4] =0 was not permitted. This was fixed.
Axis continues after reaching software limit in CSP mode.	In Cyclic Synchronous Position, when the motion arrived to a limit switch it continued the movement towards the limit if the set-point was beyond the limit. This was fixed. The motion does not continue beyond the software limits.
Unexpected move when using brake.	Enabling and disabling the motor rapidly when a brake is defined (OL[]) caused an unexpected delay during the brake engage procedure. As a result, in unbalanced systems (e.g. vertical load) the motor moved more than expected. This was fixed. The brake engaged time is as expected via the BP[] command.



Bug	Details
Quick Stop command while brake is engaged	When a DS-402 Quick Stop command is sent while the brake function is defined, and the Quick Stop Option code (0x605A) is <i>Switch On Disabled</i> (0x605A set to 0, 1 or 2), the DS-402 State (0x6041) sometimes indicated an incorrect state. This was fixed. The state is now reported as <i>Switch On Disable</i> .
Lost CAN emergency (EMCY) messages	Sometimes a CAN EMCY message, which was a result of an error code (e.g. wrong command via PDO), was not transmitted. This resulted in situations where the received command failed and the host was not aware of it. This was fixed.



Chapter 28. 01.01.10.07 B01

Component	Version
Firmware Version	01.01.10.07 05Aug2015B01
Communication Server	AA[5]: 1.1.7.77
PAL	87
ESI	0x0010400 V08

28.1 Details

Subject	Details
New G-DRUM & G-EAGLE	G-DRUM & G-EAGLE 60/100, 13/400 60/400 were added.
New G-TWITTER & G-BEE	G-TWITTER & G-BEE R15/200 were added.
New Drive: Baritone	The Baritone drive was added to the Gold family. PAL version 87 is required (VP is 87).
Motion in reset (MS=0) indication in Time depended modes	When Interpolated Position mode and Cyclic Synchronous Position mode are used, the Motion Status indication (MS) will be set to 0 when the speed command is zero. The indication is also available via the 0x1002 object (map-able) and SR command bit 18. Note that the Target reached bit in DS402 status word (0x6041) is not affected by this change and remains as defined in the relevant motion mode.
Motor Temperature measurement (PTC) in G-DUET	The G-DUET includes a motor temperature indicator. This is read via TI[5] or object 0x22A4 (map able via CAN). If the temperature exceeds 110°C, the servo will be disabled with Motor Fault (MF) 0x10000. An Emergency message with error code register 0x4210 via CAN\EtherCAT channels will be transmitted. After the servo is disabled due to the fault, it can only be enabled when temperature drops below 100°C. An attempt to enable the motor before this, will result in error code (EC) 191. Motor Temperature can be recorded by the EASII via "Motor Temperature[C]" (Group: System)
Current Integral Saturation	US[2] limits the current integral. This limit is in addition to the anti-windup limit. The range is in percentage (0..100), where 100 is max voltage output (PWM). For compatibility reasons 0 is interpreted as 100%. Limiting the integral reduces overshoots, allowing a faster settling time and maintaining integral usage for small errors.
Speed integral Saturation	US[3] limits the speed integral. This is in addition to the anti-windup limit. The range is in percentage (0..100) of the maximum



Subject	Details
	allowed current. For compatibility reasons 0 is interpreted as 100%. Limiting the integral reduces overshoots allowing a faster settling time and maintaining integral usage for small errors.
DS402 State Machine Option Code via 0x605B & 0x605C	The Shutdown option code (0x605B) and Disable operation option code (0x605C) allows the user to select the behavior of the drive when these transitions are requested via the DS402 state machine (object 0x6040). From this version, setting these objects to value 1, decelerates the motion under servo control before the servo is disabled, preventing motor spin w/o control. The deceleration that is used is the Profile deceleration (object 0x6084).
Interlock indication	The STO (safe torque off) switches are now indicated via the Interlock indication (bit 3) in object 0x60FD. The bit is set to 1 when the STO is disabled and the drive is in safe state. The indication also affects objects 0x2201 & 0x20FD.
Boot Version via EtherCAT	Object 0x200A indicates the Boot version. For example: 1.1.3.35



28.2 Bug fixes 1.1.10.07B01

Bug	Details
False Over Voltage alert	In rare occasions, the Over Voltage was falsely alerted. This was fixed.
Object 0x1000 correction	The object 0x1000 indicates the device type that is supported. In Gold the device type is DS 402. The value of the object did not indicate that the device supports servo drives. This was fixed. The value now is 0x00020192, as DS402 requires.
Uploading procedure via CAN failed to complete	The upload sequence is performed via object 0x2036 (UL command). In situations that the uploading procedure failed, the drive was locked and the CAN communication failed to respond. This was fixed.
Wrong limit indication (ILA bit) when drive encountered Software Position	The ILA bit (bit 11 of the Status Word: 0x6041) is set when internal limit occurs. This indication did not work properly when a Software position limit was reached. Prior to the fix, the ILA indication was set only if the motor was enabled outside the Position Limit boundaries. Now the indication is set whenever the software position limits are sensed. Note that the ILA indication for software limits is reset back to 0, when the motor is disabled and when mode of operation (0x6060) is changed.
Hard Stop input (IL[x]=2/3) enhancement	Hard Stop function is activated when one of the digital inputs is routed via IL[] to this function. When the input is active, the drive stops all motion. Previously, when the drive operated in time-depended motion modes (Interpolated Position mode or Cyclic Synchronous motions mode) and the Hard Stop was activated, the motion was automatically disabled. Now, the drive decelerates to full stop using the SD command and remains enabled. When the Hard Stop is active, the drive discards any motion reference and prevents any movement. This includes any motion commands such as PA, JV, ECAM, velocity and position reference via socket, as well as Yaw reference (Gantry). Note that current reference from socket will not be discarded. If the digital input is reset, the motion resumes immediately. It is recommended to use the “sticky-bit” function as described in IL[] command (Command Reference manual) and set an anti-bouncing filter via IF[] command. The FLS\RLS functionality remains unchanged.
EMCY messages via EtherCAT, wrong byte pattern	The ECMY messages are transmitted when the motor is disabled due to a fault. The ECMY that were transmitted via the EtherCAT channel included a wrong pattern on bytes. This



Bug	Details
	was fixed. The indication of the EMCY is as defined in DS301 manual.
Sync Manager 3 Watchdog	The use of SM3 watchdog caused a jump in the motion after the watchdog recovery. This was fixed. The SM3 is no longer used in the Gold drive. SM3 watchdog should be implemented by the EtherCAT host.
Gantry Yaw reference socket reset.	CA[97] denotes the Yaw reference socket when Gantry system is used. After LD command the variable was unintentionally reset to 0, instead of loading the socket number saved during SV . This reset prevented the Yaw profile from working properly. This was fixed. LD loads and initiates the reference socket.
MS behavior in Profile Torque mode	In Profile Torque mode, the Motion status (MS) was not set to 0 when the Halt command or Quick Stop state were requested via the DS402 state machine. This was fixed. Now, the MS is set to 0 whenever the motor speed is 0. Note that this does not affect the Target reached bit in DS-402 Status Word.
Battery error in Sanyo/Nikon single turn sensors.	When a multi-turn sensor was defined as a single turn and did not include a battery, a battery error (PS_ERR) was invoked. Starting from this version, in such occasions, the error is automatically reset when the sensor is initialized by the drive.
Bad reply for objects which are not in OD	Addressing an Object that is lower or higher than the lowest index or highest index of the object dictionary, caused the EtherCAT stack to lock. This was fixed. Addressing these areas in the object dictionary is now replied with "Object does not exist" abort code.
RS232 communication lost after RS	RS232 communication to the host was lost after an RS command. This happened when the communication parameters of the RS232 (parity, baud rate) were set to values other than the default. This was fixed. The RS232 communication remains after RS or LD commands.



Chapter 29. 01.01.10.05 B04

Component	Version
Firmware Version	01.01.10.05 19May2015B04
Communication Server	AA[5]: 1.1.7.73
PAL	85
ESI	0x0010400 V08

29.1 Details

Subject	Details
Panasonic incremental sensor	Version supports Panasonic incremental of 10,000 counts. Version supports Panasonic incremental 20 bits via Encoder ID 29 (CA[41] to CA[44]) . Capture function on index of this sensor was added. The capture is required for functions such as homing & touch probe.
PWM emulation error report	Feedback emulation as well as PWM emulation are set via EA[1]. If an error occurs during the initiation process, EC will be set to 107 while EE[7] or Object 0x2081.7 details the reason for the error.
UM=6, Stepper mode Open Loop, Allow modulo & Position Limit	In Stepper mode (UM=6), open loop, where no sensor is used, the modulo (XM[1]\XM[2]) and the position limit (VH[3]\VL[3]) are influencing the motion. The stepper angle is derived from the profiler output and can be used with User Units (e.g. FC[] command). The profile output in this mode can be copied to the feedback presentation (0x6064 or PU) by setting the "Copy Main Profile" (Sensor ID=15 via EAS Feedback Settings) into position socket.
Master\Slave mode for slaves without feedback	In this mode a master drive can be connected via "Gantry connection" to any number of slaves. The master transmits Current reference and field angle, which the slaves follow. The slave commutation method is set via CA[17]=7. The master commutation angle can be offset via CA[7] & CA[29], so that each slave can adjust the angle offset. Slave must be manually enabled (motor ON). The UM of all slaves should be 1 (current mode). Note that in situations where the motor ON procedure failed for the slaves, there is no indication at the Master drive. The same applies when one or all of the slaves is faulty. The Master is unaware of this situation and will continue to transmit the set



Subject	Details
	point as if all is OK. If a situation occurs where the Master is disabled, the slaves reset the current command to 0, while remaining in servo enabled.
Profile Acceleration and Deceleration range enlargement	The following parameters and objects range were modified: AC[1] to AC[4]: [2147483647...1] , setting 0 results in an error. DC[1] to DC[4]: [2147483647...1] , setting 0 results in an error. The following objects limits were also modified to [2147483647...1]: 0x6083, 0x6084, 0x60C5, 0x60C6 Note that if 0x6063 or 0x6084 is set to 0, the set point is ignored by the drive. The default values of these commands and objects were also modified.
Profile Speed range enhancements	The SP is the maximum speed of the Profile Position. Starting from this version setting 0 to SP[1] to SP[4] is permitted. If the value of SP is set to 0, the next BG command results a Stop similar to ST command. Object 0x6081 sets the Profile Velocity. Setting 0 to this object is permitted but results 'Halt' on the next setting of the New set-point bit (bit 4) of the Control Word.
Halls only speed calculation enhancement	When using 'Halls only' as the feedback sensor, the speed calculation between 2 Halls transition depends on the speed of the Halls. Previously the detection of Halls transition for the speed calculation was 1 counts every 0.1 Sec. From this version, it was improved by 20 fold, allowing calculation of 1 count every 2 Secs. This allows smoother movement at much lower speeds.
New G-HORNET	G-HORNET 9/200, 3/200, 6/200, 9/200, 12/200 were added.
Offset the stepper angle in Halls Only	CA[29] is used to move the stepper angle. From this version this feature is also permitted while the servo is on.
YM[] default value	YM[] (object 0x207B) denotes the modulo of the additional position reference for e.g. ECAM. The default value is now set to 0 which implies "No Modulo".



29.2 Bug fixes 1.1.10.05B04

Bug	Details
Cyclic Position extrapolation recovery	In CSP mode, if the next set-point failed to arrive during the interpolation period due to e.g. lost frame, an extrapolation that calculates the next set-point is used. In such situations, if the master serves as the DC clock (SYNCO) producer (slave mode synchronization), a jump in the position occurred when a valid set-point was received. This was fixed. The motion resumes the desired trajectory.
PWM saturation was not updated after US[x] command.	US[1] is used to saturate the motor PWM command (voltage command). The desired value was updated only after the relevant US[1] was sent twice. This was fixed. The saturation value is now updated immediately after sending the US[1] .
PWM Multiplication	The motor PWM frequency is limited by the current sampling time (TS) and the multiplication factor denoted by XP[2] . The drive prevents setting a combination of TS and XP[2] that might result in a frequency which can damage the drive. In some cases, modifying the TS after the XP[2] resulted in PWM frequencies which are forbidden. This was fixed. Any order for setting these commands is now permitted.
Recorder setting for Output Compare	The OC[1] & OC[2] signals can be recorded via the EAS recorder. The name of these signals is now modified to "generated pulses" instead of "remaining pulses".
XP[3] uploaded via EAS parameter	The XP[3] refer to the Interpolate mode SYNC filter and was not updated by the EAS during parameter upload and download. This was fixed.
Motor failed to move after Inhibit switch release	In rare situations, when the Inhibit function (IL[]) was released, the motion was erratic. This happened when for example a JP was requested after Inhibit is released (Enabled). This was fixed.
Drift in Home setting when User Units are used.	When User Units were used by the host (UU), the position calculated during the homing was not accurate. This was fixed. The improved calculations prevent possible drifts.
Halls only feedback, commutation angle calculations	In cases that the sensor was Halls Only the commutation angle calculation was wrong when rolling from Hall state 4 to 5. This could cause a wrong current output. This was fixed.
LD during Recorder upload caused a communication timeout	When an LD command (load data from flash) was sent during the upload of the recorder data (BH command) the drive stopped responding to the communication causing a communication timeout. This was fixed. Now during BH the LD request returns an error, EC=67 "Recorder Is Busy or Data is uploaded".



Bug	Details
Homing acceleration limit	The homing procedure was not limited by the SD (Stop deceleration) causing an ambiguity in the deceleration trajectory. This was fixed.
Emulating the speed via PWM emulation.	When speed is emulated via PWM, negative speed values were limited to -32000 cnt/sec. This limitation caused incorrect PWM signals when velocities higher than -32000 were required.



Chapter 30. 01.01.10.00 B00

Component	Version
Firmware Version	01.01.10.00 31Dec2014B00
Communication Server	AA[5]: 1.1.7.69
PAL	85
ESI	0x0010400 V07

30.1 Details

Subject	Details
Includes the following internal releases	01.01.09.11 B07 (Absolute Gantry)
Ignoring software position limit via XA[4]	When in Cyclic Position Mode, Bit 2 of XA[4] is used to ignore the FLS/RLS (forward and backwards limit switches). From this version, Bit 2 is also used to ignore the Software Position limits (VL[3]/VH[3]).
Incorrect ECAM setting new indications	When ECAM mode is initiated via EM[1], the complete set of ECAM parameters are checked and processed. If a failure occurs due to a parameters mismatch, the EM[1] command fails (EC=27) and the reason for the failure is retrieved via EE[6]. Starting from this version, the following parameters are also prevented from mismatching by the conditions: if EM[7] (last segment shorten size) > EM[4] (ECAM master gap), then EE[6] indicates 9. If EM[5] (ECAM starting index) > EM[2] (ECAM last index), then EE[6] indicates 10.
OS interpreter reply	When the OS interpreter is used, the command is sent via object 0x1023 SI 1 and the reply is performed via object 0x1023 SI 3. If after the request a SDO is sent to the drive, the OS reply will be 0. This happens when the SDO is of VISIBLE STRING type such as 0x1009, or 0x100A objects.
New mapping option to object 0x2203	The 0x2203 object namely, Application Object, can be directed to inform regarding different types of data. By default the object is used for Elmo's proprietary data. Via object 0x2F41 bits 16 to 19, namely Config Bits, the user can select the following: 0x20F41 bits 16-19 is 1: This object 0x2203 informs the analog sensor sine cosine amplitude ($\text{Sine}^2 + \text{cosine}^2$) in internal units. The expected range for 1Vp-p sensor is ~2,380,000 to 2,750,000.



Subject	Details
	0x2F41 bits 16 to 19 is 2: This object 0x2203 informs the analog sine & analog cosine signals as follows: (cosine<<16 sine) To convert the internal values to Volts, divide the value by 3085.



30.2 Bug fixes 1.1.10.00B00

Bug	Details
ESI	Some bugs related to wrong variables size, wrong defaults values, and bit offset, were fixed. These modifications will not affect any functionality or integration between EtherCAT host and the Gold drive from a previous version.
Modulo default value modified	The default value of the position limit (VL[3]\VH[3]) and the modulo (XM[1]/XM[2]) is now set to 0. Therefore, the drive will be in 32 bits modulo and the motion behavior (RADO) is “normal” similar to linear motor behavior. The default is set via the RS command.
Ignore Hardware limit switches	XA[4] sets the behavior of the limits during motion modes. The default value of XA[4] is now set to 4 and therefore the hardware limit switches (FLS, RLS & STOP) will be ignored when the drive is either in the CANopen Interpolated Position mode (IP mode), or any of the Cyclic Synchronous modes (CSP, CSV & CST).
New power rates for Gold Twitter, and Whistle	The following power rates were added to the Gold Twitter: 6/100, 1/200, 3/200, 6/200, 9/200 & 12/200 The following power rates were added to the Gold Whistle: 1/200
Homing Gantry absolute sensor	A procedure that homes the Gantry with absolute sensor was not operating properly. As part of the correction CA[110] was added. The CA[110] is an alias to OV[54] which determines the socket which is used for the DS-402 homing. Now, wherever Gantry is defined, the homing works properly and updates the relevant socket as required. NOTE: Further steps are required to perform the Gantry Absolute Homing procedure.
Transition 10 of the DS402 state machine	Commanding Quick Stop via object 0x6040 when the drive was in “Switched On” state (0x33) was ignored by the drive. This was fixed. Setting Quick Stop in this situation will set the state machine (object 0x6041) to the “Switched-on Disable” state (0x250).
CANopen LSS, ignored serial number (DS305)	LSS did not respond to the “Selected Switch” command. This occurred since the serial number of the identification was not handled properly.
Speed command exceeded VH[2] settings	Speed limit, defined by VH[2] , limited only the speed which was derived from the profiler output. This caused a higher speed than expected. From this version the VH[2] limits the total speed command to the controller which includes the profiler output of the speed derived from the external



Bug	Details
	reference (follower, ECAM, socket), and the position controller.
PR command causing an undesired modulo roll over.	When the actual position was close to the modulo limit (XM[1]/XM[2]), the next PR command caused the motor to inadvertently run to the other side of the modulo. This situation happened when a Homing (HM[1]=1) with "stop after home" (HM[4]=0) was commanded prior to the PR command. This was fixed.
Motion roll over rapidly when profile velocity is set to 0.	If the Profile Velocity (Object 0x6081) is mapped to PDO and set to 0, the profile position motion rolls over. This was fixed. Setting 0 to 0x6081 via PDO is now saturated to 1 preventing this roll over.
EA[7] to EA[14] in EAS parameter file	When uploading the parameter file via EASII, EA[7] to EA[14] were not uploaded. This was fixed.
Jog motion near the software limits	When commanding a jog motion via the JP command and the actual position was near the software position limits (VH[3]/VL[3]), in some situations (i.e. position arrived to the limit before the acceleration completed), the motor could not return to the limit as expected.
Profile Position with End Velocity near Position Limit	When a Profile Position motion was used with End Velocity and the actual position was near the Position Limits (VH[3]/VL[3]), the motion moved inside the limit. This was fixed. The motion ends at the defined limit.
Jog motion when modulo mode is active, reaches software limit	When commanding jog motion (JP) and the drive is defined as Modulo (XM[1]>VL[3] & XM[2]<VH[3]), the drive motion was limited due to the software limits (VL[3]/VH[3]). This is fixed. Now the drive ignores the software limits in such situations.
Jog when AC == SD	When commanding Jog motion (JP), if the Acceleration (AC) equals to the Stop Deceleration (SD), the motion incorporated disturbances along the trajectory.
Jog motion with various Acceleration & Deceleration	When commanding jog motion (JP) and the acceleration (AC) differed from the deceleration (DC), in some instances, the direction was incorrectly calculated causing the drive to jog in the other direction.
Incorrect Current Saturation when switching from brush to brushless motor	When switching from a Brushless to Brush motor via CA[28] , the Current Saturation (I²T) was not calculated correctly causing a lengthier peak current period than requested.
Bit 7 in DS-402 Control Word (fault reset) is Ignored	Bit 7 in the Control Word resets the drive from a Fault state according to the DS-402 State Machine (transition 15). The bit was ignored in all other States (e.g. Ready to Switch On, Switch On, Operation Enabled) causing non-requested state transitions. The bug is fixed. When bit 7 is set and the state is



Bug	Details
	not a Fault State, the transition is ignored and the drive remains in the current state.
EtherCAT Sync Manager (SM) counter modified	The SM counter is used to determine retransmission of messages. In some rare occasions if the master started the counter from 0 (for example after network reconnection), the drive lost a message. This was fixed according to ETG 1000-4 table 29. The SM handler now also responds to 0 counter value.
PWM emulation did not include Current Command	The PWM emulation (refer to explanation on PWM Emulation in the section PWM Emulation below) mode via EM[] did not include the option to emulate the Current Command (EM[9]=4) . This was fixed.
UI[] \ UF[] update via Binary Download	When uploading a parameters file using EASII via the Binary upload procedure, and then downloading the same file, the UI[] & UF[] arrays were not updated as expected. Values for all UI[] & UF[] were reset to 0. This was fixed. The values which are uploaded are now updated properly.
Switching between RM=1 and RM=0 several times caused a wrong set point	The RM command sets and resets the auxiliary reference mode. When switching between RM=0 to RM=1 several times, the profiler did not process the next set point properly. This situation is only relevant when the main profiler (DS-402 profiler) was the ECAM master via EM[11] bit 3 is set to 1. This did not occur when the ECAM master reference was from a socket (feedback follower).
PA & JP with smooth factor (SF)	When commanding Point-to-Point motion (PA) with Final Speed (FS) and smoothing factor SF , sometimes the motion moved in an unexpected direction. This occurred when the distance required for the smoothing was more than the set point calculated distance. From this version, in these situations, the motion will not move in other directions. Instead the requested Final Speed will be attained long after the target position is accomplished. In rare situations the same occurs when commanding JP (Jog Position) motion with Smooth Factor. This was fixed. The motion trajectory is always performed to the expected direction.
Motion starts when BG failed	The BG command starts a pending motion (e.g. PA , JV etc) and can be used to activate the “Begin” trigger for the recorder. Sometimes the motion starts even when the command has failed (returning error). This was fixed.



Bug	Details
PWM emulation values	The EM[14] & EM[15] parameters set the Current boundaries when PWM emulation (refer to explanation on PWM Emulation in the section PWM Emulation below) on Current is performed (EM[9]). The EA[14] & EM[15] were always loaded to Peak Current (PL[1]) value regardless of the request from the host. This was fixed. Any value can be requested and the only limit applies is: EM[14] < EM[15]
OS interpreter did not reply an error	The CANopen SO interpreter permits addressing to the Elmo interpreter via the 0x1023 SI 1 object. If the command returned an error, the user did not receive any notification. This was fixed. When a command is sent via the OS interpreter return error (EC), Object 0x1023 SI 3 reports this error.



Chapter 31. 01.01.09.12 B00

Component	Version
Firmware Version	01.01.09.12 01Nov2014B00
Communication Server	AA[5]: 1.1.7.68
PAL	85
ESI	0x0010400 V03

31.1 Details

Subject	Details
ESI new version	• Home Method modified to Read Only. • Removing irrelevant parts from the new revision



31.2 Bug fixes 1.1.9.12B00

Bug	Details
Homing attained with Modulo roll over	If during a Homing procedure the Home Attain was performed near the modulo (XM[1] , XM[2]) and the Home Offset caused the final position to roll over the modulo range, an incorrect calculation will be performed, resulting in an incorrect profiler output. This was fixed
Wrong Position (0x6064) calculation with Dual Loop	The object 0x6064 is the Actual Position value in User Units. For dual loop and when the speed sensor is more sensitive than the position sensor, the final position is calculated incorrectly. This causes an incorrect 0x6064 report to the host even if the position sensor is not actually moving. This was fixed.
Handling of Ethernet ARP message	The handling of ARP messages was performed incorrectly causing messages to be lost. This was fixed.



Chapter 32. 01.01.09.11 B07

Component	Version
Firmware Version	01.01.09.11 27Oct2014B07
Communication Server	AA[5]: 1.1.7.67
PAL	85
ESI	0x0010400 V02

32.1 Details

Subject	Details
Includes the following internal releases	01.01.09.11 B00 01.01.09.11 B01
32 bit checksum for parameters integrity	The Checksum of the parameter's database integrity was calculated in 16 bits. From this version the checksum was changed to 32 bits. OV[52] (object 0x2060) includes the checksum value. NOTE: This is compatible to previous checksum calculations. The new calculation is for presentation purposes only. All internal decisions about database integrity were not modified.
Objects 0x2110 to 0x211E	KG[] command includes 15 filters. Each filter includes 63 parameters. These filter parameters are indicated by 945 sub-indices. Objects 0x2110 to 0x211E are used as alias to the KG[] where each filter is represented by a different object. For example: Object 0x2110 includes 63 parameters of the Speed KI filter entries and 0x2111 includes 63 parameters of the Speed KP filter. This allows a CANopen or EtherCAT host to address all the filters via an SDO (CoE) interface.
EtherCAT stack revision number (ESI)	The EtherCAT revision number was modified to 0x0010400.
Position Error indication for Gantry Yaw & Stepper Closed loop	The Position Error of gantry Yaw controller or Stepper controller is determined via ER[5] . If the position error is violated, MF will be set to 0x400 (1024) and an EMCY message is transmitted. Previously when the fault occurred, MF was set to 0x100.
Auto Focus error bit indication	The Auto Focus is Elmo's proprietary additional sensor that can be read via the drive inputs. Bits 8 & 10 of object 0x2085 are reserved for this special mode.



Subject	Details
US[1] modified in motor enable	The US[1] sets the saturation for the PWM (voltage command of the current controller). The value can now be modified even when the motor is enabled.
Dynamic brake enhancement	The Dynamic Brake (DB) reduces the coasting time of the motor after it was disabled. By shortening the lower phases when the free-wheeling speed is lower than VH[1], the coasting time is reduced. Previously this speed was checked only during the motor off interval. From this version, the threshold speed is checked always and not only during the procedure of the motor off. This means that even if the motor is not enabled, the DB will be activated when actual speed $\leq$ VH[1].
Dynamic brake with no sensor	Where there is no speed sensor, e.g. UM=1 with Brush Motor and the VH[1]>0 , the motor enable request (MO=1) will return the Error Code 80.
Adding new power rated for Twitter (GTWI)	Following power rates are added to the GTWI: 6/100 1/200, 3/200, 6/200, 9/200, 12/200
Adding new power rate for Whistle (GWHI)	1/200 module is added.



32.2 Bug fixes 1.1.9.11B07

Bug	Details
Reset the PAL during RS command	The PAL was not reset to its default after the RS command. In rare occasions this caused an absolute sensor initialization issue if no power up cycles were performed after the RS .
Motor off when changing motor type via CA[28]	CA[28] determines the motor type. This cannot be modified during motor enable. This restriction and bug correction is applicable also for the stepper mode.
Position behavior on Stepper Closed loop (UM=6)	In stepper closed loop, the profiler output was not synchronized to the sensor at motor on and the position limits were not set correctly. The user units were also not calculated correctly. This was fixed.
Inhibit/Enable wrong operation	When a digital input was modified from Inhibit/Enable function (via IL[]) switches to any other function (e.g. general purpose) the motor was incorrectly enabled. This caused an unexpected behavior since the function was not defined as Enabled any more. This was fixed.
Phase D lost synchronization with Gantry	When using Gantry with Stepper 2-Phases motor, the current of phase D was losing synchronization. This was fixed.
Object 0x2020.1 out of range	The 0x2020.1 is used during the home on block settings. If a 32 bit long value (e.g. 0x2F00.1) was sent to the drive prior to the 0x2020.1, the drive sometimes replied with an "Out of range" abort message. This was fixed.
Home attained jump while using position FIR filter	The position FIR filter is set via CA[88] . When the filter was activated (i.e. CA[88]>0), the position jumped during Homing. This was fixed.
Absolute offset when analog sensor is used (CA[91] to CA[94])	The CA[9x] is used for absolute sensors offset. An analog sensor is also defined as an Absolute sensor within the single Analog cycle. When an analog sensor is used, the offset during the Homing procedure was calculated from the absolute position (PX) instead from the analog signal. This caused, in some occasions, a wrong offset calculation and as a result a wrong homing position. This was fixed.
Wrong default value for 0x606A	Object 0x606A is the sensor selection object that determines from where the velocity is read. The user can select between the speed of the position sensor and the speed of the velocity sensor. The default was set to 0, meaning the position sensor was read. This was fixed. The default is now set to 1, which indicates that the velocity sensor is read.



Chapter 33. 01.01.09.10 B03

Component	Version
Firmware Version	01.01.09.10 12Aug2014B03
Communication Server	AA[5]: 1.1.7.61
PAL	85
ESI	0x000103F6 V07

33.1 Details

Subject	Details
Allowing parameters reset while features are active	The RS command resets the drive parameters to their factory default. The RS is now available when the following features are active: • Output Compare (OC command), • Position Error (PC command) • Emulation (EM command) NOTE: The feature will be disabled when RS is activated.
Emulation Saturation works as gain as well	EA[16] & EA[17] are used to saturate the PWM emulation signals. From this version it can be used additionally as gain. Therefore when the PWM emulation exceeds EA[12], then EA[16] is used. When the PWM exceeds EA[13], EA[17] is used. When the PWM emulates the Velocity then EA[16] and EA[17] are used instead of EA[14] and EA[15].
Gold Clarinet Drive	Introduces the Gold Clarinet drive which is similar to the Gold Bell stepper drive. The Power rates are similar to the Gold Bell.
Disabling the Profile Acceleration limiter (XA[4])	The acceleration (and deceleration) limiter denoted by the SD command, prevents rapid speed changes. In time depended modes, for example Cyclic Position, Interpolated Position and when using object 0x2005 where the set-point reference is controlled by the network master, the acceleration limiter is automatically ignored allowing the host to set any required acceleration. In other modes which are not time depended such as Profile Position mode, the acceleration limiter can be override by setting bit 1 of XA[4] command.
Ignoring Hardware limits (FLS, RLS & hardware Stop)	When the motion mode is time dependant, the hardware limits can be ignored by setting bit 2 of XA[4]. The hardware limit override is relevant when 0x6060 is set to modes 7, 8, 9 and 10. In all other modes the hardware limits cannot be ignored.



Subject	Details
Absolute sensor CRC error	If a serial sensor is used and a CRC error is reported, the remaining status data is ignored. This modification prevents extraneous status info, because the sensor data is no longer valid.
Specific serial and incremental sensor errors must be manually cleared	If a Serial Sensor reports an error, the motor is automatically disabled. The next motor on will not be allowed until the error is cleared via the TW[20] command (Object 0x20FC sub index 2). The value must be set to the relevant sensor socket e.g. TW[20]=1 , resets the errors of sensor in socket 1. The relevant sensors are: EnDat 2.2, Panasonic Absolute, Panasonic Incremental, Tamagawa, Sanyo/Nikon
Battery warning indication via 0x2085	Object 0x2085 bit 6 will be set if the sensor reported a battery warning alert.
Masking errors indications of specific serial and incremental sensors	CA[8] can be used to mask status bits of the following sensors: Panasonic Absolute, Panasonic Incremental, Tamagawa, Sanyo. Setting CA[8] to -1 masks all the errors
Gurley sensor validation bit	The Gurley sensor must be moved before the position data is validated. When Bit 9 of object 0x2085 (OV[61]) is set to 1 , it indicates that the Gurley sensor data is valid.
Warning bit of DS-402 Status word indicates threshold exceeded	Extra Status register (Object 0x2085 or OV[61]) indicates when these thresholds of selected data were violated. The threshold are set via the XT[] command. If bit 2 to 5 or bit 6 in 0x2085 is set, than bit 7 ("Warning") of the DS-402 status register (0x6041) is automatically set.
PWM Follower new indication	WS[39] indicates the PWM follower Duty Cycle /150 in MHz WS[40] indicates the PWM follower Period Time /150 in MHz WS[41] indicates the Duty Cycle in %
PWM Follower value when signal is faulty.	When the PWM follower signal is out of range and considered as "cable disconnected" (Duty Cycle 0% or 100%), the actual PWM by default will be set to 0. This can be modified via CA[109] . CA[109] is defined in units of mili ampere, or cnt/sec depending on the CA[101] index that selects the PWM type.
G-TWI & G-BEE increased threshold temperature	The threshold temperature for G-TWI & G-BEE products with max current (MC) <=6 was increased from 85° to 92°.
Current Step shaping	Enhanced current step shaping was added to improve the current control step response.



33.2 Bug fixes 1.1.9.10B03

Bug	Details
0x30xx objects with extended indices	The 0x30xx objects are aliases to the Elmo's legacy Commands. For these objects, sub-indices which were higher than 254 could not be addressed. This was fixed. All the indice range are applicable to the host.
Sine\Cosine synchronization	In Port-B when the sine / cosine sensor is used, the drive rolls over at very high speeds (>3e6 cnt/sec). This was fixed.
Peak Continuance current limit at motor off do not reset the filter	At motor off (MO=0) the filter of the current limit saturation (CL[]\PL[]) was reset. In this situation, the current limit protection (I²C) did not work properly. This was fixed. The filter of the current protection is now active regardless whether the motor is enabled or not.
Start Position of Output compare function is not detected (first pulse)	The monitoring of the first pulse for the Output Compare mode is not detected, if the difference between the hardware counter (Quadrature module) and the user position (PX) is larger than 2 ³¹ .
Limited acceleration for the field angle	When the difference between two commutation cycles was greater than 60 electrical degrees (Rate limiter), the field angle was saturated. This limit was removed, allowing a faster commutation response. Note: When Hall sensors are used for commutation, an error is produced when the difference between two commutation cycles was greater than 60 electrical degrees.
JP[4] was not allowed if motor is disabled	JP[4] sets the jog reference to the socket profiler. During motor off (MO=0) this setting was rejected. This was fixed. The socket profiler can be activated regardless of the MO status.
FLS/RLS limits the wrong speed command	FLS & RLS limited the speed derived from the software command. The speed derived from the external reference was not limited. This is now fixed. FLS & RLS limits are now working on the superimposed values of the software and external speeds.
RM=0 causing wrong set-point calculations	When switching from RM=1 to RM=0 , the external command was ignored, causing the reference signal to skip. This was fixed.
Jog Position in negative direction	JP[2] , JP[3] and JP[4] are used to set a jog reference to the superimpose, phasing and socket profilers. In these modes, when the command was negative, the profiler ignored the value. This was fixed. A negative value can now be set.



Bug	Details
Long motion with low acceleration using simple profiler	When using superimpose, phasing or socket profiler with a lengthy motion ($> 2^{31}$ bits) and low acceleration, the position calculation lost counts. This was fixed.
Interpolation Profiler vibrations	In rare situations the Interpolated Position mode vibrates causing small interruptions in the motion. This was fixed. This did not affect any other motion modes.
Speed residue when FIR filter used	When a Velocity FIR filter was used, the speed residue was not correctly cleared causing continuation of slow movement regardless of the set-point. This was fixed.
Wrong state indication in DS-402 status word	Object 0x6041 indicates the Status of the DS-402 state machine. In some occasions when transitions 3 and 4 were used at the same cycle (switching from “Ready-to-Switch-On” to “Operational Enable”), the Status word indicated 0x235 which is an illegal state. This was fixed.
OV[61] wrong indication	The OV[61] indicates the Extra Threshold register (object 0x2085). In rare situations, a negative value was displayed. This was fixed.
0x30xx objects with extended indices	The 0x30xx objects are alias to the Elmo’s legacy Commands. In these objects, sub-indices which were higher than 254 could not be addressed. This was fixed. All indices ranges are applicable to the host.
Hiperface sensor was not read correctly	This was fixed with PAL 85.



Chapter 34. 01.01.09.08 B08

Component	Version
Firmware Version	01.01.09.08 12Jun2014B08
Communication Server	AA[5]: 1.1.7.57
PAL	82
ESI	0x00103F6 V07

34.1 Details

Subject	Details
Absolute sensor daisy chain	If GO[14] to GO[16] are set to 8, then the absolute sensor is buffered to PORT C (daisy chain) as follows: Port C 'A': Buffer clock out of the sensor Port C 'B': Buffer Data input of the sensor Port C 'Index': Buffer Data output of the sensor
Extended frequency for SSI sensors	CA[36] in addition to the present clock frequencies for the Absolute Sensor, now allows the frequency of the SSI sensor to be set to 312KHz
Polarity bit modified (object 0x607E)	Previously, the polarity bit of the DS-402 (object 0x607E) modified the direction of the velocity or position commands to the relevant controller. Now, due to an expected change in the DS-402 standard, and in order to allow modifying the polarity value while motor is enabled, the polarity affects the set-point instead of the command.
Stepper Commutation for unbalanced system with and without mechanical brake.	The stepper commutation is now also capable of commutating in unbalanced systems where for example the gravitation affects the load. The total time allowed for the commutation process is now set to 20 seconds. New variables were added to the SC[] command for this purpose. Note that SC[4] and SC[9] are now reserved. Any setting to SC[4] will result in 0 and any setting to SC[9] will result 1. This is for compatibility reasons. The new commutation procedure is also possible in commutating against a block where movement is limited or not permitted at all.
Commutation verification for Binary search	The user can now request to run verification after Binary Commutation. This is done via the SC[12] command.



Subject	Details
During Stepper commutation drive returns to the initial position.	When drive is in Position Mode after the Stepper commutation is complete, it returns to the initial position. Note that in Velocity and in Current modes the drive remains in the final position after the commutation and does not return to the initial position.
Stepper Closed Loop	Stepper closed loop mode via UM=6. The sensor ID is 34. This includes the following: • The Position command source is now set via CA[98] • Feedback socket is set via CA[99] • The relevant controllers are KP[4,5], KI[4], KV[81..90] and FF[2] • KP[6] converts from encoder counts to stepper angle units (512 per period) • CA[53] determines the source of stepper angle to this socket. The Stepper closed loop can also be used to close any sensor for emulation purposes.
XP[8] : Improved Gantry Master Slave synchronization	The XP[8] command has been introduced for the Phase Offset to improve the synchronization in gantry systems between the master and slave.
PWM reference as input to the drive	A PWM reference can be used via socket. The sensor ID is 35. CA[100] to CA[108] are used to determine the Duty Cycle factors and relevant parameters for this mode. The mode includes protection for slow amplitude and protection from cable disconnection (0% or 100% duty).
PWM Emulation	The Emulation function that emulates signals to an external host via PORT C can now be used with PWM duty cycle. Pulse & Direction or +/-50% modes can be used. The PWM can be set with a Gain & Offset. New EA[] command entries are used to program the desired functions of this feature. WS[11] informs the duty cycle. TW[70] can be used to manually set the PWM value mainly for debug purposes.
Sanyo & Nikon feedback reset	Single-turn & Multi-turn position resets are now available with these sensors via the commands TW[18] and TW[19] (Objects 0x20FC SI 1, and 0x20FC SI 2) respectively.



Subject	Details
"Motor failed to start" error is reported via EC command	When the motor is enabled via MO=1 or DS-402 and failed to start, the abort code is reported as: "Motor failed to start" via MF command (0x20000000). The reason for the failure is now reported via EC command. The EE[5] and Object 0x2081.5 reporting this abort previously, are not in use and always reports 0.
Inhibit Function and RM command	The Inhibit function is not dependent on setting the RM anymore. Previously RM=1 was required. From this version when the Input is defined as "Inhibit/Enable" it will enable and disable the motor regardless of the RM value.
DD command reports the drive CAN controller status	The DD command reports the drive CAN controller state indication; BUS off, Errors counter, error type and more. This can be used to monitor the state of the CAN bus. Object 0x2082 can also be used for this purpose. NOTE: In a situation of BUS off the CAN Bus is no longer active, and can only be monitored via the DD command.
Jump in motion In Cyclic Position mode (CSP)	If the communication was interrupted during Cyclic Position Mode the drive extrapolates the motion, when communication is returned (host sent new set point) and reconnected, the position was calculated wrongly. This occurred when, in the course of the disconnection, the extrapolated position passed the modulo point.
Homing direction for Home On Block modified	The homing direction for home on block is modified: -1 positive direction -2 negative direction
Adding Serial number to the USB 3	For USB 3 compliant tests, a serial number was set during the enumeration stage. Note that this is causing the Windows System manager to allocate a different COM port than was used previously.
BOOT Strap LED indications	Adding a green blinking LED when the EtherCAT BOOT strap state is requested. LED is toggled in 200Hz (50mSec).
EtherCAT switches lo added for Gold TWI/BEE	EtherCAT switches is allowed with Gold TWI\BEE drives in all hardware configurations of the drives.



34.2 Bug fixes 1.1.9.8B08

Bug	Details
Motor Enabled when EtherCAT state is in SafeOP	The motor can be enabled by various methods. When in EtherCAT field bus, the servo is enabled via the DS-402 Control Word (0x6040). This typically occurs when the EtherCAT state is in OP (Operational) mode. If in these situations, MO=1 command is sent via other communication channels (e.g. USB or RS232), the command will be overridden by the object 0x6040. When the EtherCAT state is in SafeOP, the MO=1 command was ignored. This was fixed.
Final Speed (FS) negative value	The FS command accepted negative values. This was fixed. Positive values are only permitted.
CAN baud of 1M – sampling method	The SAM bit of the CAN network was modified from 3 sampling points to 1 as requested by the CANopen standard.
XA[2] was wrongly calculated	In a high resolution sensor the hot-plug speed threshold (XA[2]) allowed enabling of the motor where it should be prevented. This was fixed.
UM=3 user unit	In Unit mode 3, the reference command (e.g. PA) was calculated with User units. This was fixed, and User units are ignored when UM=3.
Sine amplitude 0 allowed movement	When the Sine amplitude was set to 0, the profiler ignored any other requested movement. This was fixed.
JP command not working when switching modes	When switching from Torque or Velocity mode to JP mode (Jog under position mode), the mode switch did not operate correctly, resulting in no jog movement. This was fixed.
Wrong Position presented after PX exceeds 31 bits	In some situations, when the PX exceeds 31 bits the calculation of the presented position (PU) was wrong. This effected only the representation via the recorder but not the motion. This was fixed.
Use CA[80] & CA[81] only if not 0	CA[80] defines the socket for the Gantry slave and CA[81] for the Gantry master. The use of these parameters is only valid, if the value is not set to 0.
Profile Torque switch – torque drop to 0	When switching to the profile torque mode the first torque command dropped to 0. This was fixed. The first torque command is now derived from the user set-point via object 0x6071.



Bug	Details
Cyclic Position Mode recovery	Wherever the set-point in CPS mode was not received, the drive extrapolated the last command according to the speed and acceleration. If the extrapolated position switched 32 bits, the next target position was not calculated correctly, causing a jump in the motion.
Hall commutation in 1:1:1 mode	When Hall commutation was used with Halls only feedback and the control mode (XA[5]) was set to 1:1:1, an incorrect commutation was calculated.
OS interpreter	When the OS interpreter produced an error which resulted in an EC, the '?' were set to the SDO data segment causing a redundant byte in the OS response.
Stop By Switch with wrong deceleration	In Cyclic Sync Position mode (CSP) and Interpolated Position modes the Stop-by-Switch caused a deceleration according to Quick Stop without taking into account the SD . This was fixed. The deceleration is now calculated by the lower value between Quick Stop and SD .
Objects 0x22A2 & 0x22A3 sub 1 mapping	These objects, indicate the temperature of the drive, are only map-able to a tPDO via CAN.
Digital input auto routine event are missed	Digital Inputs can be read via the AUTO routine programmed by using the User Program. The AUTO routine is programmed to react when the desired DIN is set to '1' (e.g. AUTO_I1 for digital input 1). Occasionally the AUTO routine missed an Input event. This was fixed.
EC command missed from EAS report	The EC reports errors that occur during command interpretation. Some of the EC were missed from the EAS personality which resulted in an "unknown error" report. This was fixed.
PC[3] default value modified	PC[3] determines the table which will be used for the Position Error correction. The default value was set to the NF table, is now switched to ET[] allowing 2046 entries.
CANopen Heartbeat correction	When the Node ID for the Heartbeat consumer is set to 0, the function is no longer activated. Previously, the heartbeat consumer was still reacting to messages causing a wrong EMCY messages if heartbeat was activated. The heartbeat starts only if the time is defined via 0x1016 is > 0 and the node ID is a legal node address (1 to 127). Otherwise the mode is not active.
IP mode caused a crash	When switching to Interpolated Position mode the drive may crash. This was fixed.
Wrong mapping causing a crash	In rare situations when an object was wrongly mapped to a PDO, the drive may crash. This was fixed. In such cases an Abort message will be transmitted and the EC will be set to 31.



Bug	Details
Undesired movement in Stepper modes (UM=3)	In stepper mode, the profiler did not start from 0 causing an undesired movement. This was fixed. The profiler sets 0 at the first cycle.
ECAM output jumps	If the ECAM output passed the 32 bits. A jump occurred.
EtherCAT ID switches after firmware update	After firmware update via FoE the ID switches were not read correctly. This was fixed.
Hot plug speed calculation for serial sensors	The speed for the hot-plug (XA[2]) was wrongly calculated, preventing the drive to servo enable even at low speed.
Wrong checksum calculations for the non-volatile parameters	Object 0x2060 reports the checksum of the non-volatile parameters. In some occasions the calculated checksum included variables which were not part of the non-volatile parameters. This resulted in an incorrect checksum. This was fixed.
FoE firmware downloading failure	In rare situations the FoE procedure caused an internal loss of synchronization which resulted in a timeout. This was fixed.
Wrong WS[30] reported	In some situations WS[30] which reports the drive type was wrong. This affected mainly the Gold TWI drive.



Chapter 35. 01.01.09.04 B01

Component	Version
Firmware Version	01.01.09.04 07May2014B01
Communication Server	AA[5]: 1.1.7.53
PAL	81
ESI	0x00103F6 V07

35.1 Details

Subject	Details
Adding Gold TIGER to the Gold family	The Gold Tiger was added to the Gold family.
USB 3 ACK	Acknowledging the end-point is required by the USB 3 (was not required by USB 2).
KL[2] to terminate user program	KL command (or KL[1]) terminates the User Program and disables the motor. KL[2] terminates the User Program while leaving the motor enabled (if enabled before the command)

35.2 Bug fixes 1.1.9.4B01

Bug	Details
EMCY was transmitted once	After a failure, an emergency message via CANopen or EtherCAT was transmitted only once. This was fixed. If the event occurred again the EMCY will be retransmitted.
EMCY was not sent during motor on failure	The EMCY message is also sent when the motor enabled failed. If during motor enabling a fault is detected, the expected EMCY message was not transmitted. This was fixed.
Wrong asynchronous PDO event reaction	CANopen PDO can be transmitted in various events as defined by object 0x2F20. The "Home Attained" event was wrongly switched with the "Auxiliary Home" event and the "Motion Complete" was wrongly switched with the "Motion Started" event.



Chapter 36. 01.01.09.00 B00

Component	Version
Firmware Version	01.01.09.00 09Mar2014B00
Communication Server	AA[5]: 1.1.7.51
PAL	81
ESI	0x00103F6 V07

36.1 Details

Subject	Details
Includes the following internal release	01.01.08.02 B00
Fault reaction for feedback faults	The Fault reaction option code (0x605E) allows the user to determine the behavior of the motion when a fault occurs. If the fault is related to feedback (e.g. Digital hall sensor, commutation) the drive aborts immediately regardless of the selected option code in Object 0x605E.
XA[3] default value modification	The Default value of XA[3] was modified to 600 µSec. For firmware compatibility reasons setting the XA[3] to 0 will automatically be modified to 600.
STO version change	Working with Dynamic brake caused the STO diagnostic to fail. This was fixed. Starting from this version the VS is modified to: 1.0.12.0



36.2 Bug fixes 1.1.9.0B00

Bug	Details
Position indication jumped around hardware modulo	When the PX exceeds 31 bits, the calculation of Absolute Position PU or Object 0x6064, and the Position Demand (0x6062) was incorrect. This was fixed
Calculation of the Position in non-position modes	A wrong position calculation in Profile Velocity and Profile Torque caused a jump when switching between these modes to profile Position. This was fixed.
Modulo when setting PX	When the user sets the PX value, the actual value is calculated with modulo. These calculations were sometimes wrong. This was fixed. PU (0x6064) and PX are now reported correctly.
Communication lost during upload personality with CAN	When uploading personality with CAN drives and TS > 60, the communication was sometimes interrupted. This was fixed.
Output compare undesired pulse	When setting OC[1]=1 and OC[1]=0 sequentially, sometimes a pulse was generated by the drive. This was fixed.
Control Word recording delay	The DS-402 Control Word (0x6060) arriving from the host can be recorded by EASII. The recorded signals represent a delayed value from the value that is handled by the drive. This was fixed. The actual Control word handled is now also recorded.
Jump in speed when switching modes	When switching to Profile Position mode, the previous velocity is used instead of the Profile Velocity. This caused a jump in speed after a few servo cycles. This was fixed.



Chapter 37. 01.01.08.00 B00

Component	Version
Firmware Version	01.01.08.00 13Feb2014B00
Communication Server	AA[5]: 1.1.7.8
PAL	81
ESI	0x00103F6 V07

37.1 Details

Subject	Details
Includes the following internal release	01.01.07.20B01 01.01.07.19B01 01.01.07.19B00 01.01.07.17B00 01.01.07.16B08
Differences between SR register and CAN status register (0x1002)	The SR is reflected in object 0x1002 and should provide the same value at the same cycle. The bug was that the 0x1002 was updated a cycle after the SR . This was fixed. Values are updated at the same cycle.
Motion continues when switching from profile velocity or profile torque to profile position	When switching motion modes (0x6060) from profile velocity (or profile torque) to profile position, the drive continues to coast according to the last Velocity set-point. This was fixed. The drive now decelerates when switching modes, using Quick Stop Deceleration.
AN[2] read wrongly via EtherCAT	The object 0x2205.2 which reflects AN[2] reported wrong value. This was fixed. The value of AN[1] & 0x2205 SI 2 are similar.
AN[2] : Analog input 2 indication	The analog input 2 can be retrieved via the AN[2] command. The expected values are from 0 to 4095 which are correlated to 0 to 3V. An offset can be used via AS[2] . This Offset has a range from -4096 to + 4095.
XA[3] : Current exceeded time protection	This variable sets the protection where the current exceeds the peak limit range but has not reached the “short” threshold. When the current is within this range for XA[3] time, the motor will be disabled. For compatibility reason setting XA[3] to 0 will automatically be changed to the default 600 µSec.
XA[2] : Hot plug In protection	When the motor is freewheeling at high speed, enabling the motor (namely hot-plug) may damage the drive.



Subject	Details
	XA[2] limits the freewheeling speed which is allowed when enabled. By default the value is set to 5, indicating that when the servo is enabled, the motor speed must be lower than 5 electrical angles/sec, for the servo to be enabled.
Gold Tuba drive	The Gold Tuba drive was added
Includes the following internal release	01.01.07.16B08
Gold Bassoon new power rated version	Bassoon 3/200 was added
Elmo commands via SDO interface (CoE)	The 0x30xx object allows addressing of Elmo Legacy Commands. This interface is now open for all Elmo users. This is done by using the two letters legacy method to determine the command offset starting from object 0x3000.
Asymmetric current limits via 0x2005	Object 0x2005 can be used to address various functionalities of the drive. By setting GS[12]=3, the host can set asymmetric torque limits to the current controller. The low 16 bits are used for the low limits and the high 16 bits to the high limit. The limits affect the current command derived from the velocity controllers to the current controller. Units are 1/1000 of rated torque.
Changing default of IL[6] & IL[7]	Digital Inputs default was modified to "Ignore" instead of General purpose.



37.2 Bug fixes 1.1.8.0B00

Bug	Details
EMCY messages are not transmitted	The Object 0x2F21 is used for the EMCY mask when CANopen is employed. The Object was initialized properly causing the EtherCAT interface to block all EMCY. This was fixed.
Wrong "In Torque" calculation	The "target reached" and the MS command value in Profile Torque mode were calculated based on an incorrect variable. This was fixed. The calculation is now based on the current feedback.
Target Torque missed resolution	The integer calculations of the Target Torque were truncated to the lower value, sometimes causing a wrong set-point to the Current profiler.
FF[4] : phase advance in 1:1:1 mode	The phase advance was not working properly when switching to 1:1:1 mode (XA[5]). This was fixed.
Touch Probe position	When using touch probe the reported position did not consider the User Unit factor (e.g. FC[]). This was fixed.
Halls indication in EASII do not match the recorder indication	The Halls are indicated via the SR command. The EASII recordings of the Halls differ from the SR indication. This bug was fixed. The indications are now identical.
Motion did not fault when serial feedback disconnected	When absolute feedback was disconnected from the drive, an incorrect reading of velocity caused a very high speed indication which was not correlated to the actual speed. This was fixed by ignoring the reading in cases of CRC and by reading the filtered speed. This behavior is only relevant when the Fault Option Code object 0x605E is set to 1 or 2 (Stop to 0 speed and then disable).
Position Offset in Homing	During Home Attain, the final Homing position differed from the desired position offset.
Quick Stop command wrongly handled	During the Quick Stop request the initial position for the profiler stop is calculated. This calculation is sometimes incorrect causing undesired motor movement. This was fixed.
STO error during drive power up	After powering up the drive, when the Dynamic Brake is used (VH[1]>0), the STO diagnostic procedure indicates a failure in the STO mechanism.



Chapter 38. 01.01.07.15 B00

Component	Version
Firmware Version	01.01.07.15 01Jan2014B00
Communication Server	AA[5]: 1.1.7.5
PAL	80
ESI	Elmo ECAT DriveGcon_v70

38.1 Details

Subject	Details
	01.01.07.14B14 01.01.07.14B08 01.01.07.14B00
EMCY mask object modification	Object 0x2F20 is used to mask unrequired EMCY messages for the CANopen. Bit 1: is now defined as “mask motor on EMCY” and will prevent the EMCY when the Motor On fails. Bit 2: is now defined as “mask un-configured PDO EMCY” and will prevent EMCY messages when the drive receives rPDO which are not mapped.
Phase advance to stepper closed loop	Phase advance (FF[4]) allows increasing the speed by advancing the stepper angle relative to the motor speed. This mode is now applicable to UM=6 open loop as well. The phase is advanced relative to the velocity command.
Improving the speed response in Stepper Modes	The speed modification of the stepper mode was limited to 30 electrical degrees in a single cycle. This was now improved to 60 electrical degrees, allowing faster response.
Resetting absolute position	TW[18] command is used in the Absolute Sensor to reset the single turn absolute position reading to a desired value. For: • Tamagawa & Panasonic sensors only 0 is permitted • EnDAT2.2 any value is permitted • Setting others sensors will result in an error code (EC=99)
SSI 2 via digital output 3, 4	SSI sensor can be read via digital inputs 3 and 4 as single ended sensor (sensor ID: 33). This configuration allowed the handling of 3 sensors in parallel. Note that a single absolute sensor can be read by the drive



Subject	Details
Auto Focus mode	The Auto Focus is an Elmo proprietary additional sensor that can be read via the drive digital inputs If error occurs when using this special mode, then bit 8 in object 0x2085 will be set.
JP[4] is enabled at Motor Off	JP[4] sets a jogging trajectory to the socket profiler. The socket profiler can be activated regardless whether the motor is enabled or disabled. Setting JP[4]=x ; BG[4] causes the socket profiler to jog. Note: more variables are required for socket profiler activation.
Addition of Gold Falcon drive	The Gold Falcon drive was added to the Gold line.
Nikon sensor support 20 bits	The Nikon sensor single turn now supports 20 bits. This was added to the 16 bits multi turn resulting in a 36 bits sensor (20 + 16)

38.2 Bug fixes 1.1.7.15B00

Bug	Details
Floating point resolution	Some fractional values were not calculated correctly, e.g. 0.99999 resulted in 0.1 instead of 1. Now the fractional values has been improved to correct the results.
Quick stop transition from Fault state	When the drive is in fault state and a Quick Stop is requested, the state indication was wrong. This was fixed. The drive ignores requests to Quick Stop when in Fault state.
Target reached (MS=0) after mode of operation switch	When switching from any mode to cyclic mode, the target reached indicated a wrong value. MS reported 0 where 2 is expected. This was fixed.
User Program size limited to 16K	The user program size is now enabled to 32K.
Set BG in motor off modified the Motion Status	Setting BG command when the motor is disabled modified the MS from 3 to another value. The MS value must remain 3, when the motor is disabled. This was fixed. NOTE: When motor is disabled, the Begin Motion trigger in EASII can be activated via the BG .
Double Torque in CST mode	In Cyclic Torque Mode (CST) the torque command is doubled relative to the Torque set point. The torque is now corrected.



Bug	Details
Object 0x6074 (Torque Demand value) returns wrong value	The Torque Demand reported a value to the Current Controller derived only from the profiler output. This was fixed. The Torque Demand now reports the summation of all the Torque inputs e.g. Torque external reference.
DV[2] reported wrong speed command	DV[2] reported a value to the Velocity Controller derived only from the profiler output. This was fixed. DV[2] now reports the summation of all the Velocity inputs e.g. Velocity external reference.
Interpolated Position Mode	When the IP active bit was set to 0 by the host the motion continued. This was fixed. Drive now stops the motion.
Motion persisted when switching Profile Velocity to Position mode	When switching from Profile Velocity to Profile Position mode, the motion continued with the Profile set point.
NRZ sensors do not transmit EMCY when in error	If an NRZ sensor e.g. Panasonic sensor, is in error state, the motor enable command fails without a proper emergency message. This was fixed. A message is now transmitted in such occasions.
In Cyclic Torque Mode first torque is always 0	When switching to Cyclic Torque mode the initial cycle uses zero torque set point, causing the torque command to drop its value regardless of the set point (object 0x6071). This was fixed and the initial cycle now uses the desired set point.
In 1:1:1 mode (XA[5]) calculates the incorrect speed	When using the 1:1:1 control mode the velocity is calculated incorrectly. This was fixed.
Default IP address	The IP address default values were reported incorrectly This was fixed. The default values are now shown as follows: IP: 192.168.1.49 (AA[10]) Subnet mask: 255.255.255.0 (AA[11]) Default GW: 192.168.1.1 0 (AA[12])
0x1003 sub index 0 is static	Object 0x1003 reports up to the last 16 EMCY messages. Sub index 0 of this object is used to indicate the number of EMCY that were reported. The value was set to 16 regardless of the actual number. This was fixed.



Bug	Details
0x603F was not updated via ECAT interfaces	Object 0x603F reports the last EMCY. The object was not updated correctly nor transmitted to the host. This was fixed.
Current continued when the torque command is 0	When using acceleration feed forward (FF[1]), sometimes the control loop received a torque command without having any target torque. This was fixed.
Sine signal launching (TW[80])	When sine movement is initialized, and both signals of the sine are 0, the profiler continues the motion at constant speed. This was fixed.
Change SD during MO=1 double caused a jump	When SD is changed during Motor On, an incorrect deceleration is calculated, causing a jump in motion. This was fixed.
Analog sensor jumped when LD is requested	After the LD command is requested, if the sensor is analog sine/cosine sensor or Hiperface sensor, the following motor enable request will cause the motor to jump. This was fixed.
Hiperface position reading after LD is requested	When the Hiperface sensor is used, and the LD command is requested, the position reported is incorrect. This was fixed.
HL[2] negative values	HL[2] parameter sets the threshold to the velocity feedback. Negative values were permitted. This was fixed.
ST[N] did not work properly	The ST[N] is use to stop the N profiler. The N refers to the Main, Superimpose, and phasing or socket profilers (1 to 4 respectively). Previously ST stopped all four profilers. This was corrected so that ST[4] stops the socket profiler and ST[1] or ST stops the other profilers.
Wrong behavior at Low speed and High acceleration.	Excluding the main profiler, the other profilers; Superimpose, Phasing and socket profilers behave erratically when at low speed and high acceleration. This was fixed.



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